I2200: Digital Image processing

Lecture 3: Grey and Color Image Processing

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Feb. 11, 2026

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Some materials from Dr. Lexing Xie , Alan Peters, and Dr. Shahram Ebadollahi

Outline:

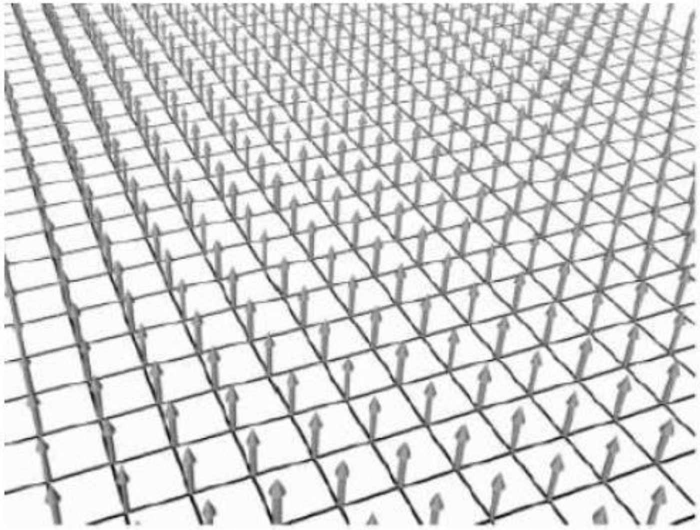
- Preliminaries
  - Sampling function
  - Fourier Transform
- Image Histogram
- Color
  - Visual Perception
  - Color Representation
  - Color Models and Transformations
  - Color Sampling and Interpolation

Preliminaries

Delta function: $\delta(x) = 0, x \neq 0$ $\qquad \lim_{\varepsilon \to 0} \int_{-\varepsilon}^{\varepsilon} \delta(x) dx = 1$

Sampling function: $comb(x) = \sum_{n=-\infty}^{\infty} \delta(x - n\Delta x)$

$$s(x, y) = \sum_{j} \sum_{k} \delta(x - j\Delta x, y - k\Delta y)$$



Preliminaries – Fourier Transform

- The Fourier Transform is an important image processing tool which transfer an image from the spatial domain to the *Fourier* or frequency domain.
- Application:
    - Image filtering
    - Image reconstruction
    - Image compression

Preliminaries – Fourier Transform

1-D FT: $$f(t) \rightarrow F(\omega) = \int_{-\infty}^{\infty} f(t)\exp(-j2\omega t)dt$$

1-D Inverse FT: $$F(\omega) \rightarrow f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\exp(j2\omega t)d\omega$$

2-D FT: $$f(x,y) \rightarrow F(\omega_x,\omega_y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\exp[-j(\omega_x x+\omega_y y)]dxdy$$

2-D Inverse FT: $$F(\omega_x,\omega_y) \rightarrow f(x,y) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(\omega_x,\omega_y)\exp[j(\omega_x x+\omega_y y)]d\omega_x d\omega_y$$

http://en.wikipedia.org/wiki/Fourier_transform

Discrete Fourier Transform (DFT)

- The DFT is the sampled Fourier Transform and therefore does not contain all frequencies forming an image, but only a set of samples which is large enough to fully describe the spatial domain image.
- The number of frequencies corresponds to the number of pixels in the spatial domain image, *i.e.* the image in the spatial and Fourier domain are of the same size.

Discrete Fourier Transform (DFT)

For a square image of size N×N, the two-dimensional DFT is given by:

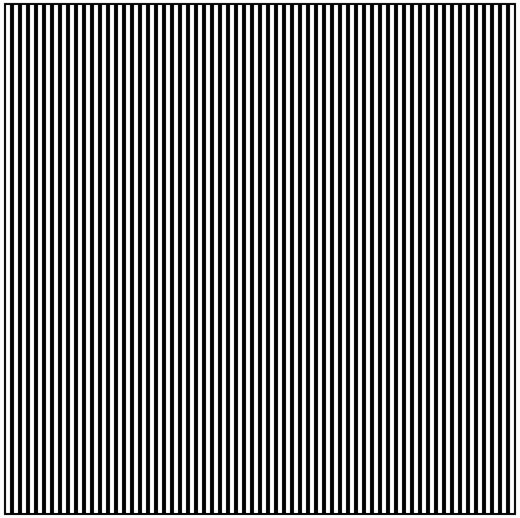
$$F(k,l) = \frac{1}{N^2} \sum_{a=0}^{N-1} \sum_{b=0}^{N-1} f(a,b)\, e^{-\iota 2\pi \left(\frac{ka}{N} + \frac{lb}{N}\right)}$$

f(a,b) is the image in the spatial domain and the exponential term is the basis function corresponding to each point *F(k,l)* in the Fourier space.

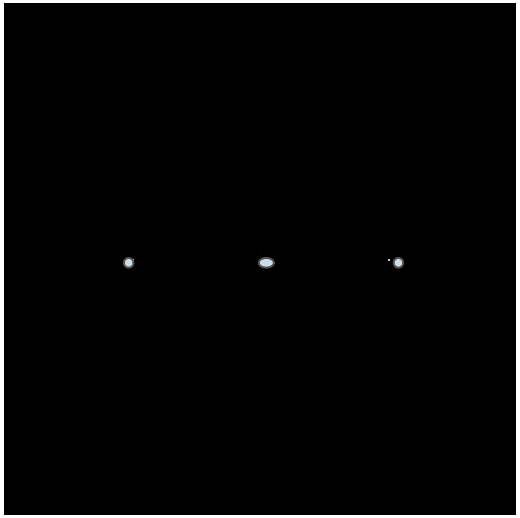
Inverse DFT:

$$f(a,b) = \frac{1}{N^2} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} F(k,l)\, e^{\iota 2\pi \left(\frac{ka}{N} + \frac{lb}{N}\right)}$$

Fourier Transform Pair - example



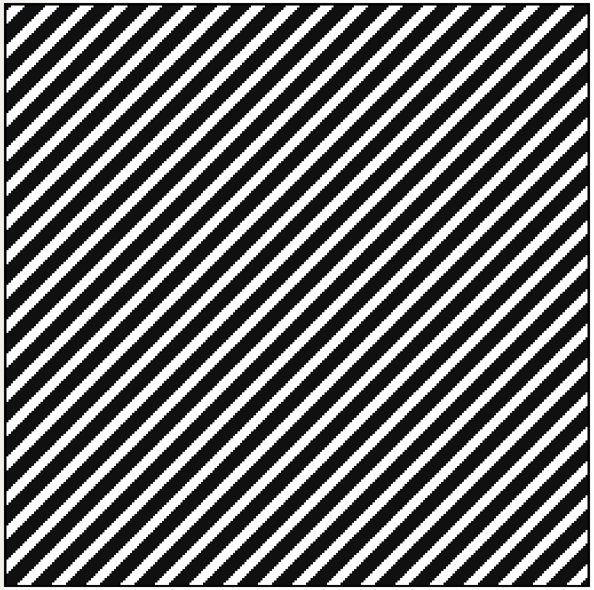
Image



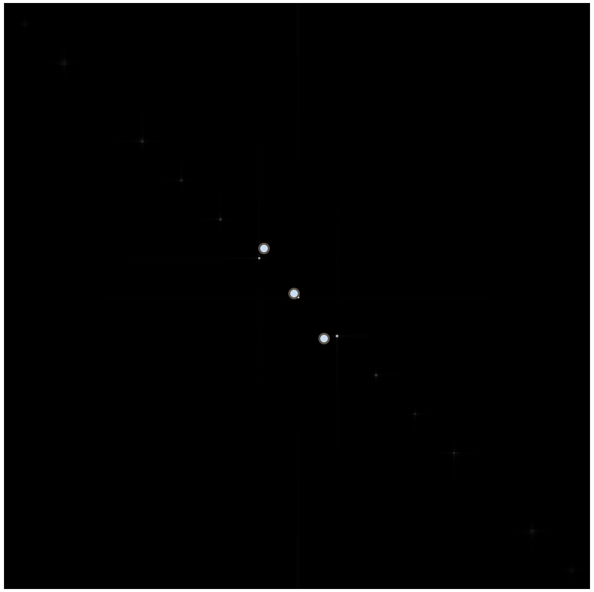
Fourier Transform

To learn more:
http://homepages.inf.ed.ac.uk/rbf/HIPR2/fourier.htm

Fourier Transform Pair - example



Image



Fourier Transform

To learn more:

http://homepages.inf.ed.ac.uk/rbf/HIPR2/fourier.htm

Fourier Transform Pair - example

Image

Fourier Transform

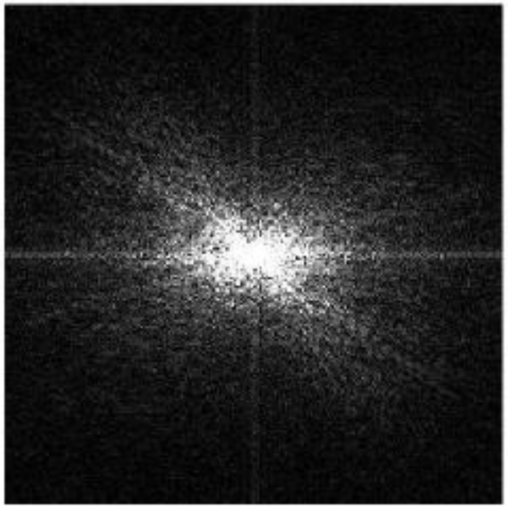


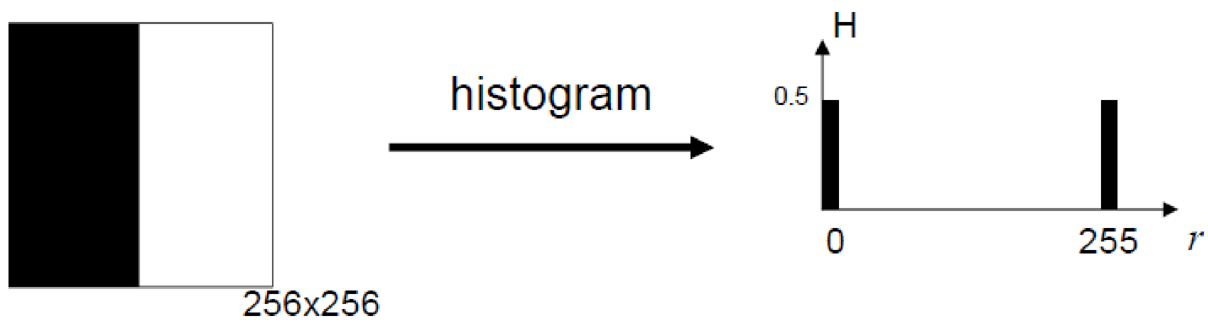
Image Histogram

$f_{MxN} \quad , f(m,n) \in \{r_0, r_1, \cdots, r_k, \cdots, r_{L-1}\}$

$$H = (h(0), h(1), \cdots, h(k), \cdots, h(L-1)) \quad , h(k) = \frac{n_k}{MxN} = \frac{\sum_m \sum_n \delta(f(m,n) - r_k)}{MxN}$$

normalized

bi-level image



Pixel values linearly increasing from 0 to 255 with increasing column index

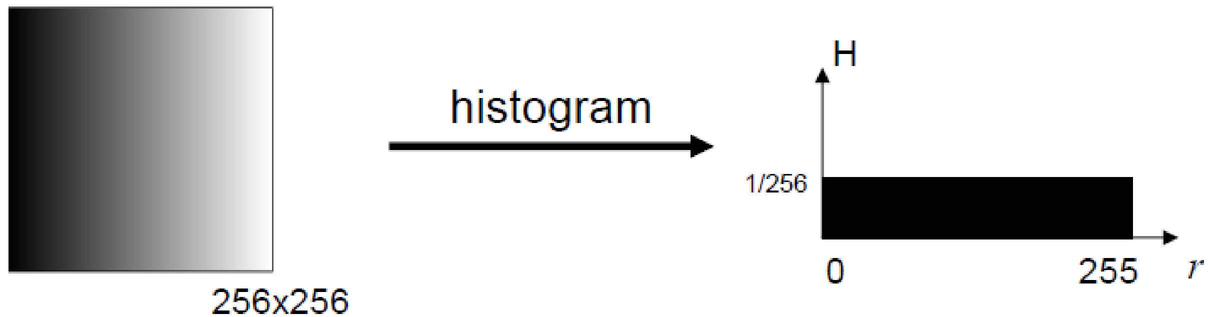


Image Histogram: example

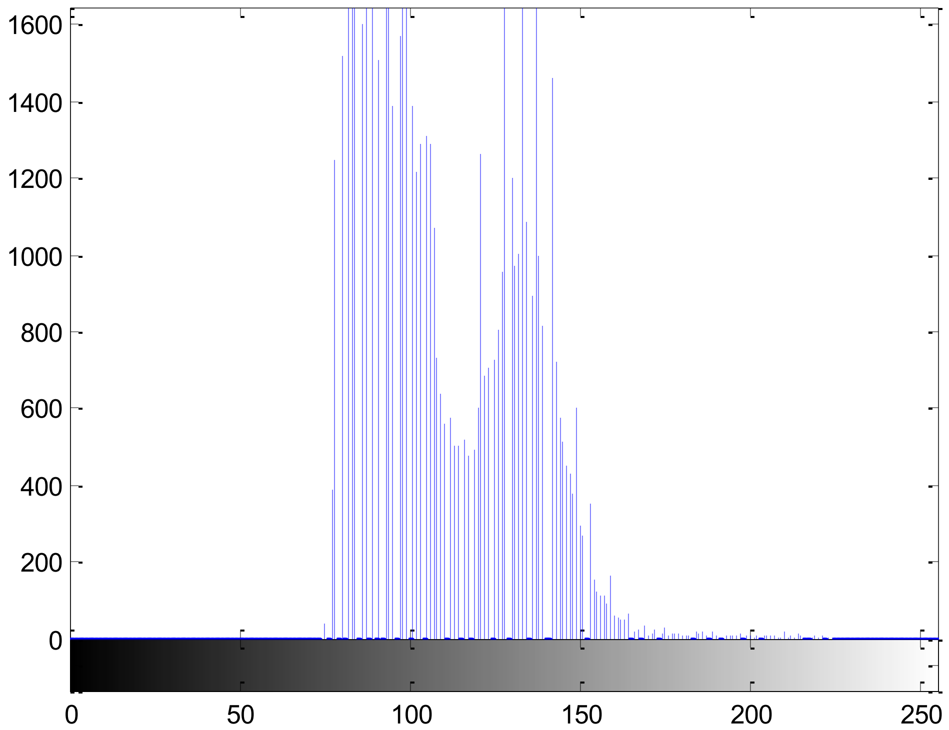
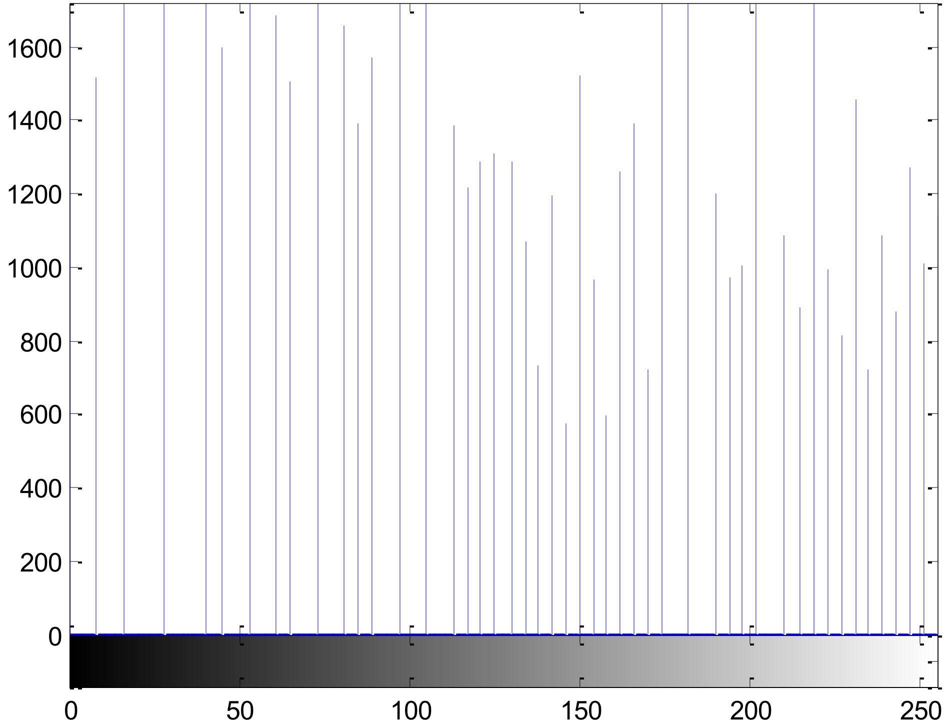
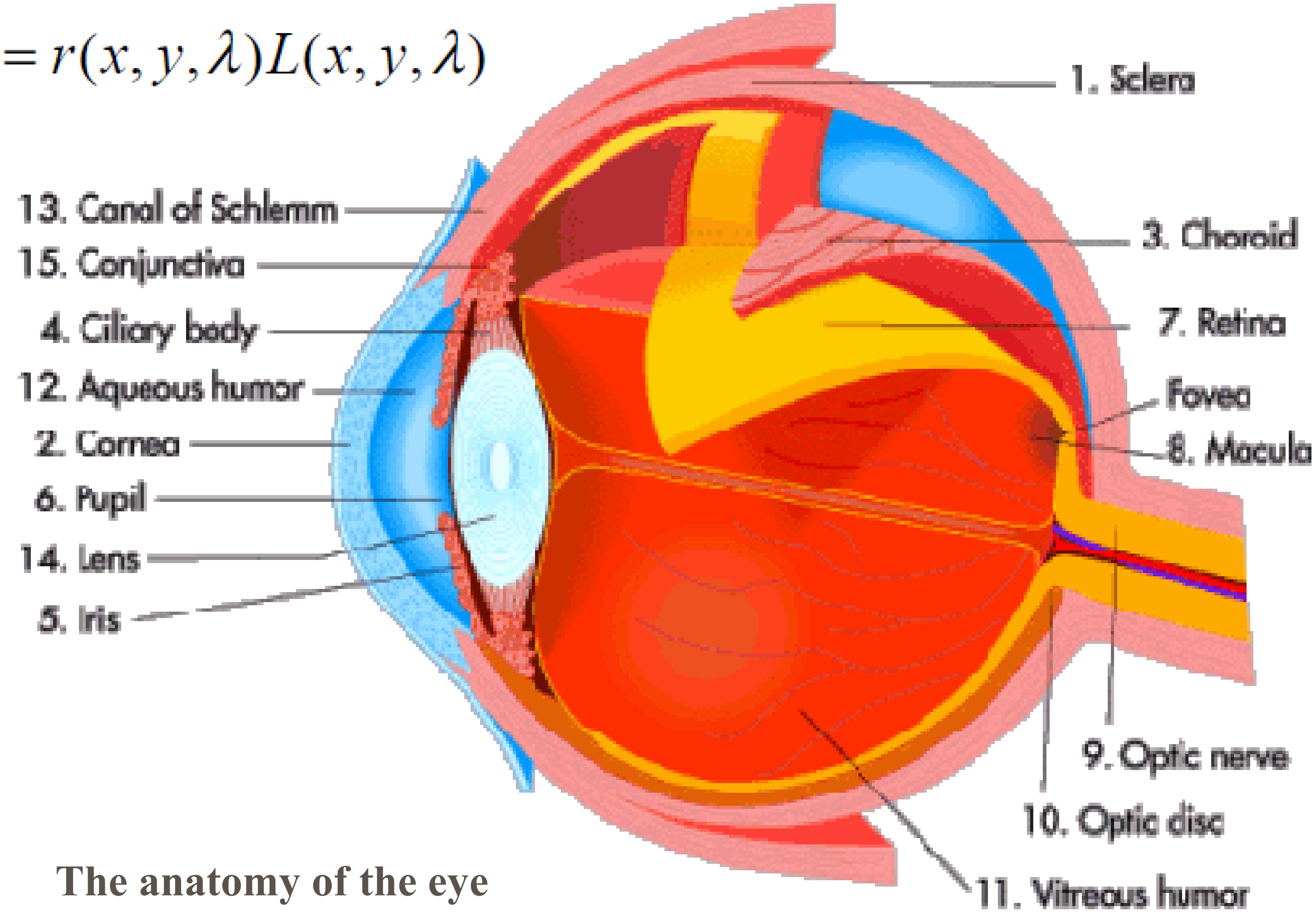


Image Histogram: example



Eye Physiology & Visual Perception

$$I(x,y,\lambda) = r(x,y,\lambda)L(x,y,\lambda)$$



The anatomy of the eye

Image © HowStuffWorks.com

How Human Eye Works?

- Scattered light from the object enters through the cornea.
- The light is projected onto the retina.
- The retina sends messages to the brain through the optic nerve.
- The brain interprets what the object is.

How Your Retina Works?

Its main function is to receive and transmit images to the brain. These are three main types of cells:

- Rods: able to function in low light create black-and-white images without much light
- Cones: able to see **color** and detail of objects with enough light.
- ganglion cells: interpret the messages from the rods and cones and send the information on to the brain by way of the optic nerve.

The Retina

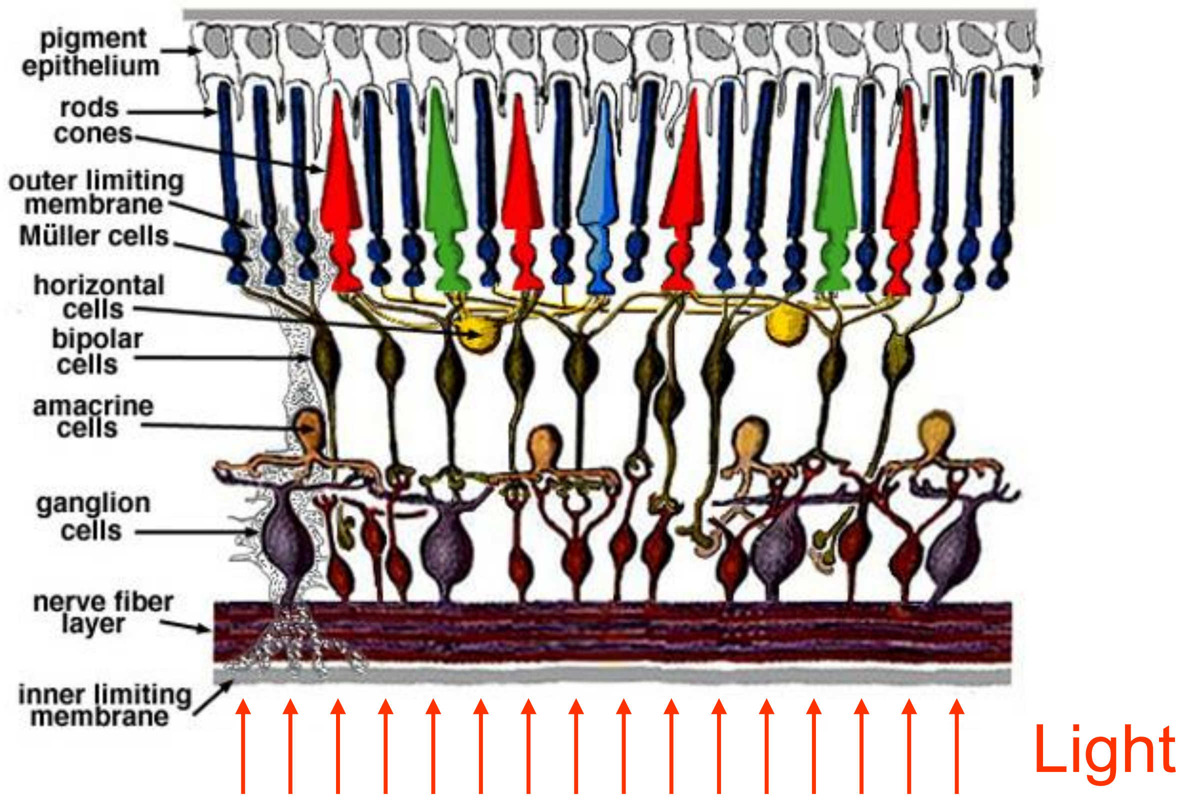


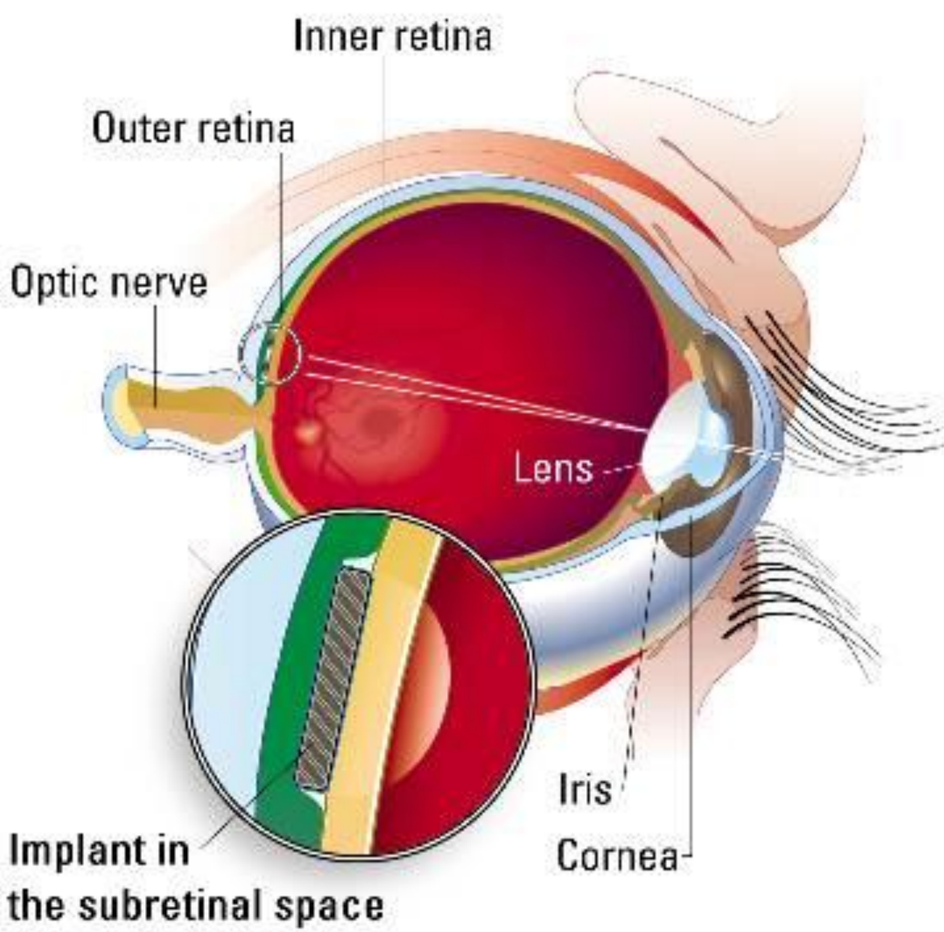
Fig. 2. Simple diagram of the organization of the retina.

Questions?

- What is near sighted?
- What is far sighted?
- What is the meaning of 20/20 vision? (20/40 vision)?
- Blind or Visually impaired

Artificial Silicon Retina (ASR) for the blind

diameter 2 mm and thinner than a human hair



Here you can see where the ASR is placed between the outer and inner retinal layers.

Image from http://health.howstuffworks.com/artificial-vision.htm

Color Image Processing

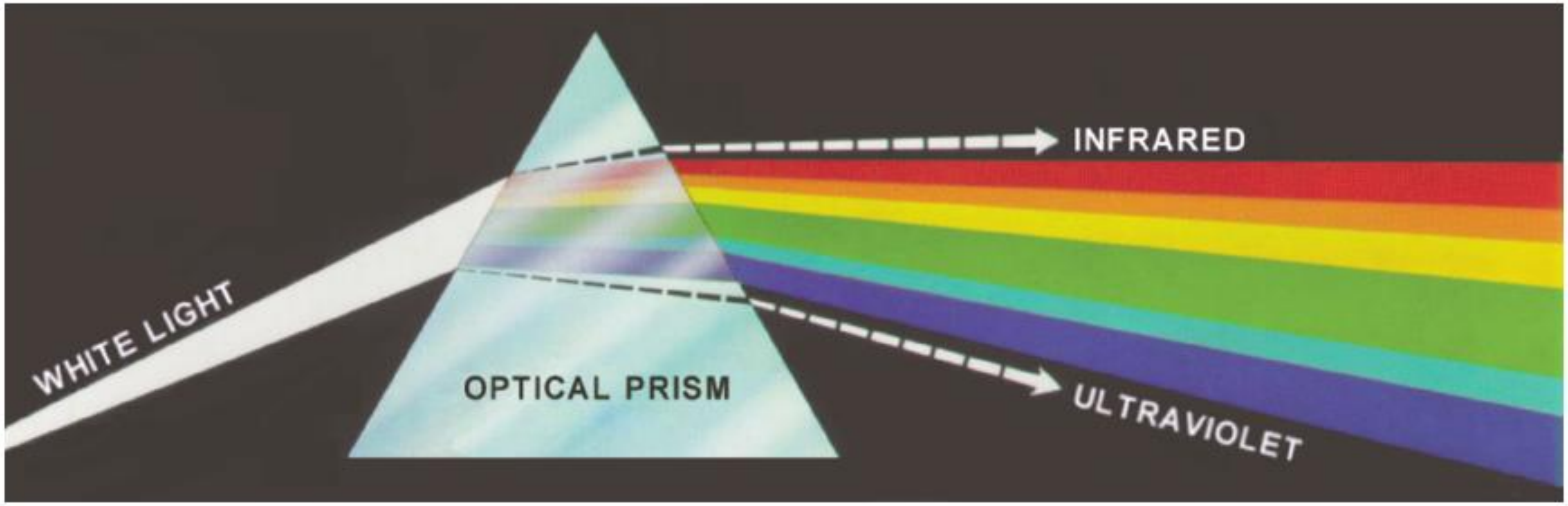


FIGURE 6.1 Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)

Violet, blue, green, yellow, orange, and red

Color Image Processing

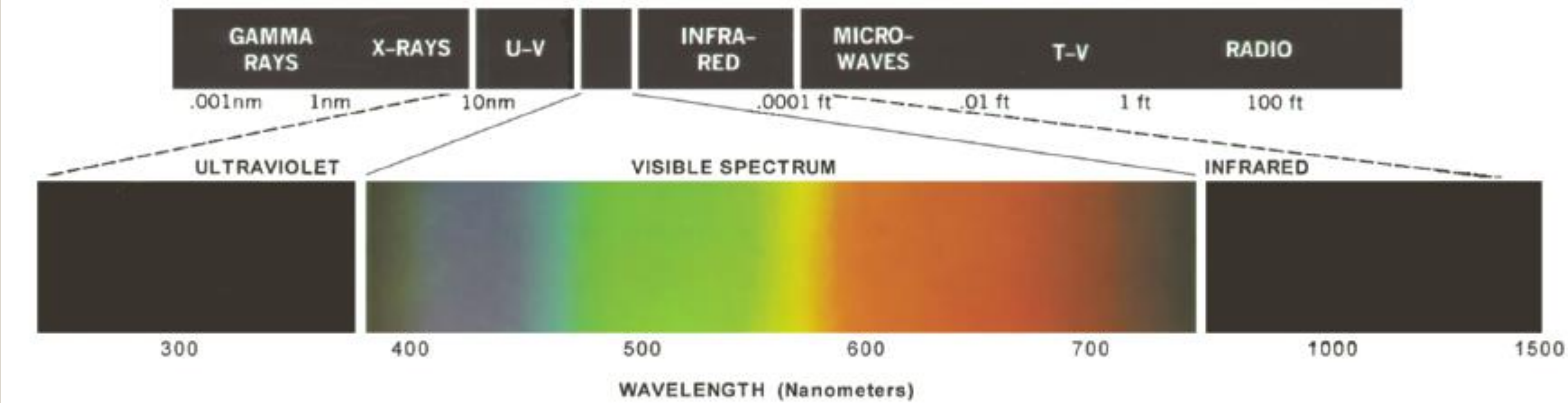


FIGURE 6.2 Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

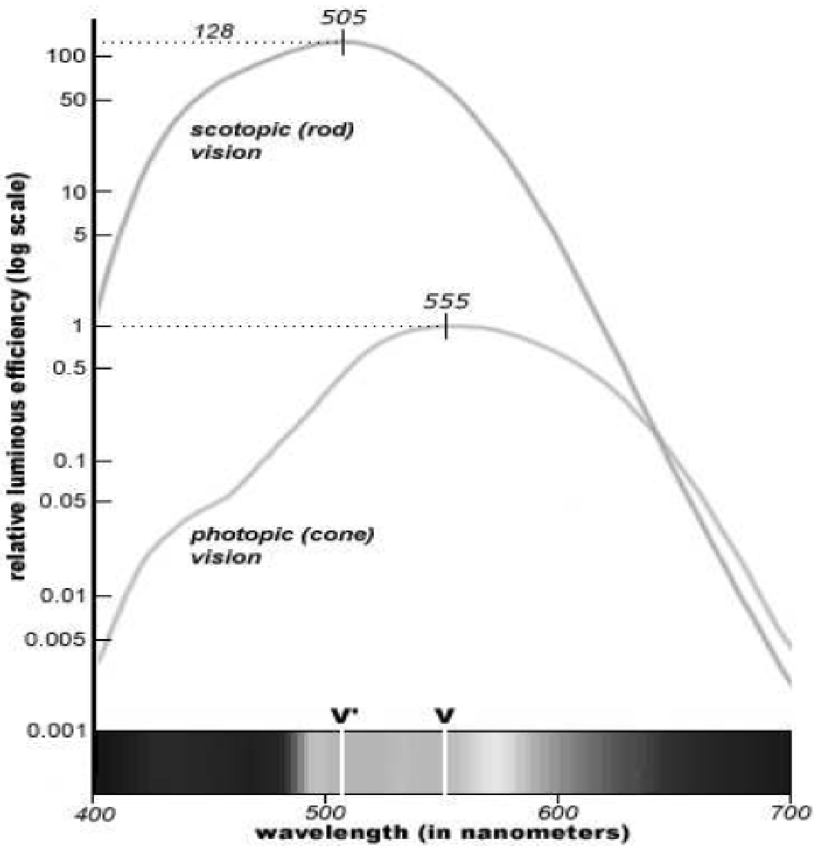
Visual Perception: Luminance

Spectral Power Distribution of the Stimulus

$$I(x,y,\lambda) = r(x,y,\lambda)L(x,y,\lambda)$$

$$F(x,y) = \int_0^\infty I(x,y,\lambda)V(\lambda)d\lambda$$

Luminance (intensity)



Radiance and Luminance

- Radiance: total amount of energy that flows from the light source.

  Unit: watts (W).
- Luminance: the amount of energy an observer perceives from a light source.

  Unit: lumens (lm)

Visual Perception: Color

- Humans can perceive 100-200 gray levels but thousands of colors

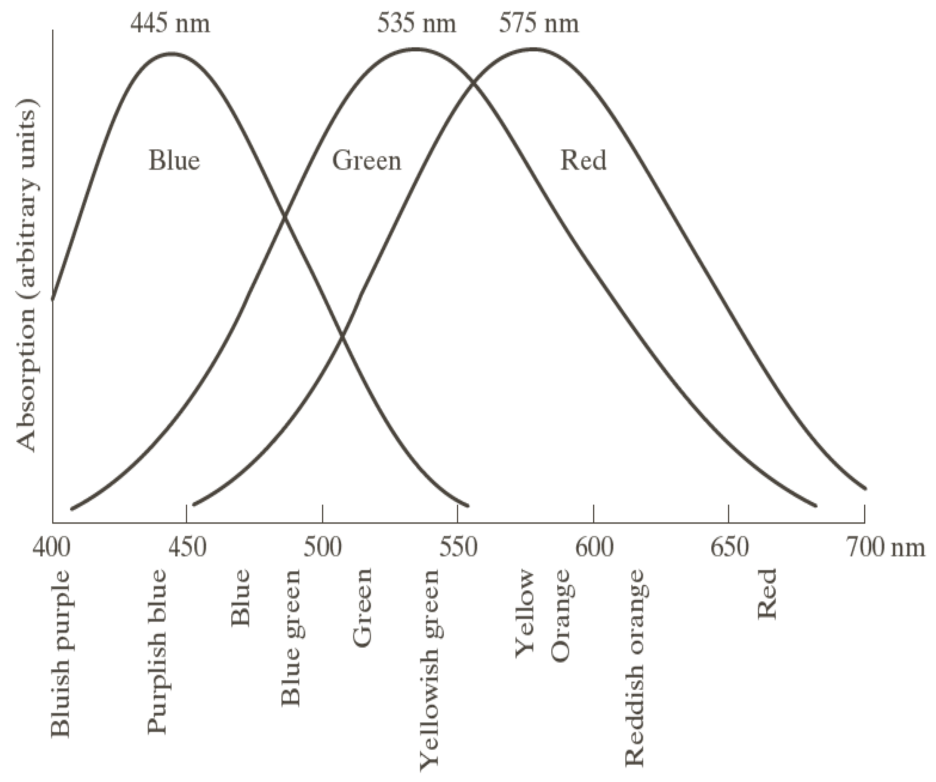
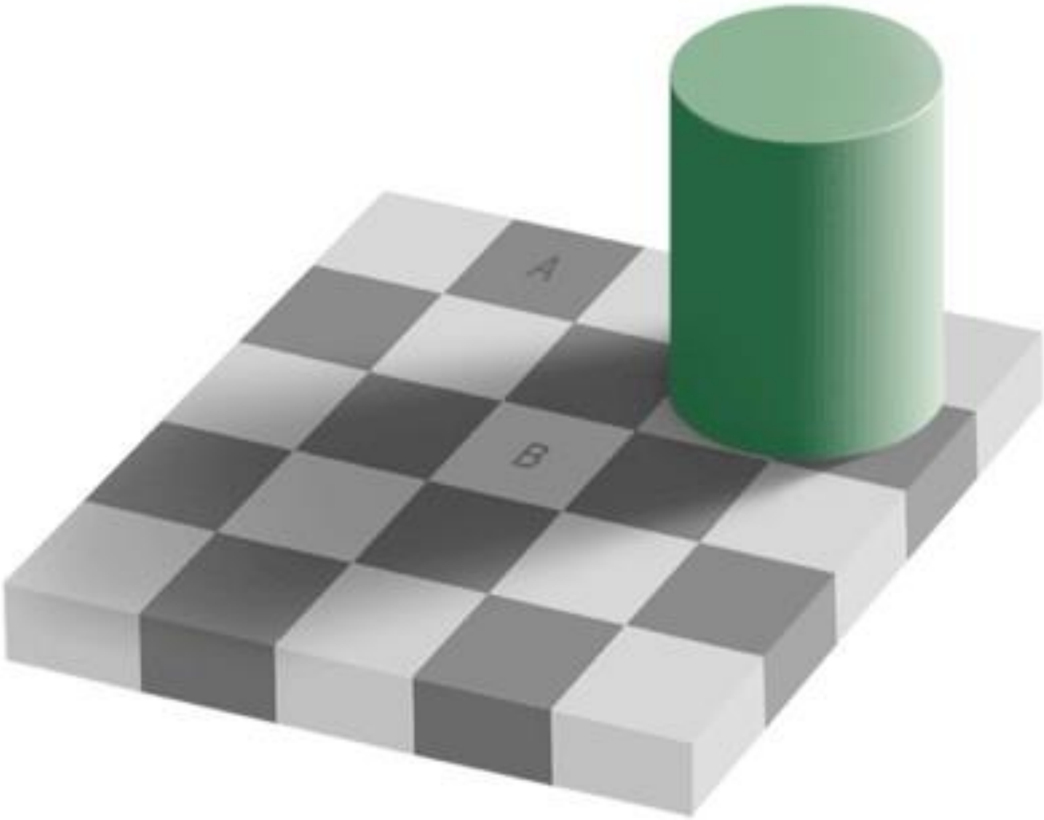
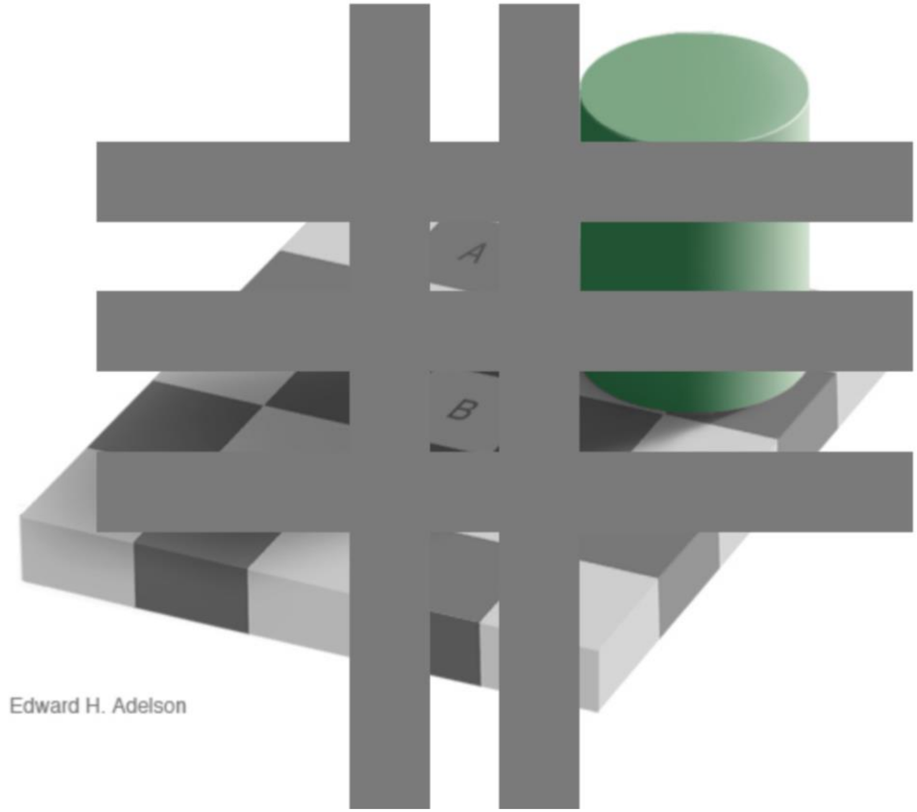


FIGURE 6.3 Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

Color Illusions

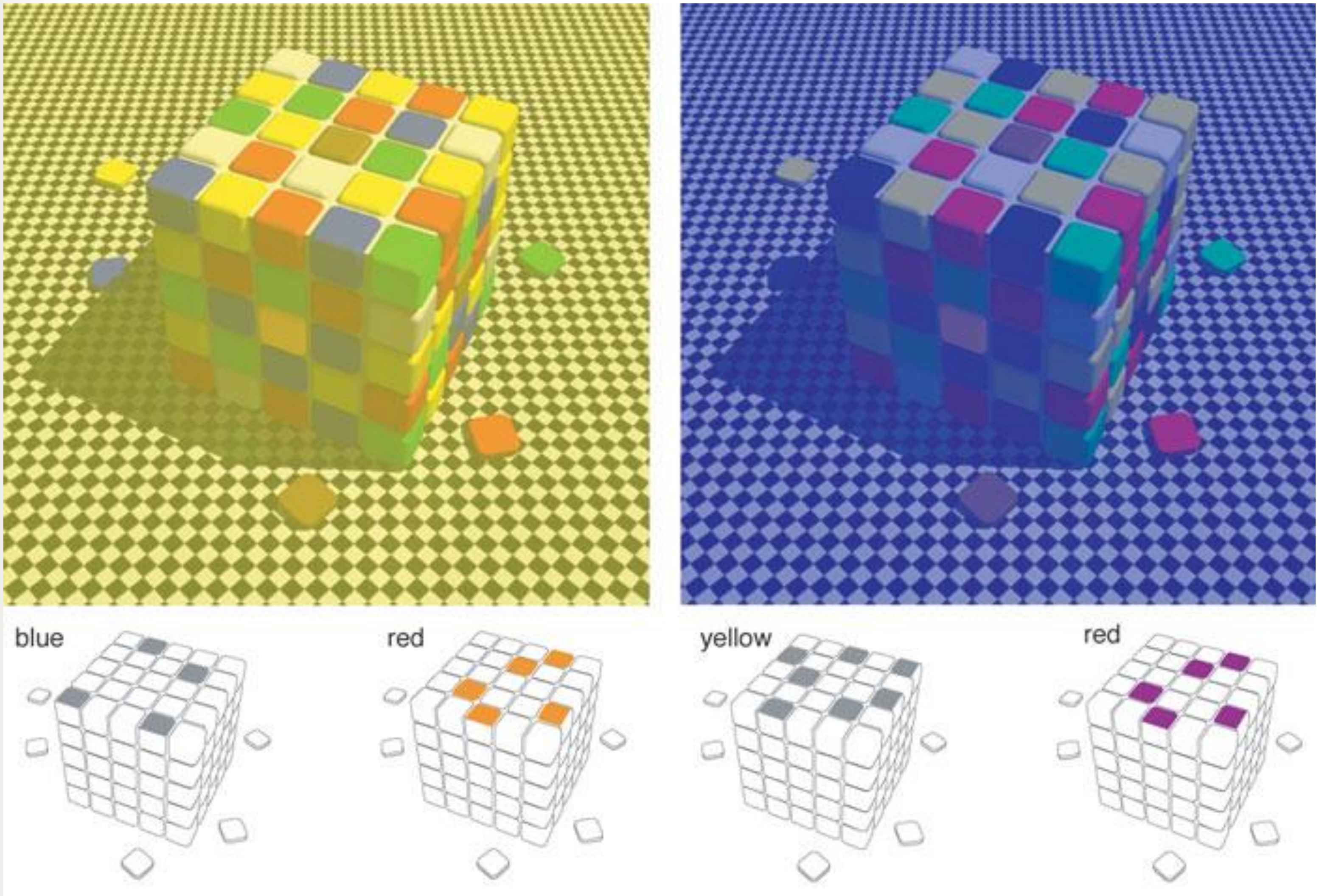


Color Illusions



Squares A and B are the same shade of gray ((It was created by Edward H. Adelson, Professor of Vision Science at MIT.)

Color Illusions



yellowish illumination bluish illumination

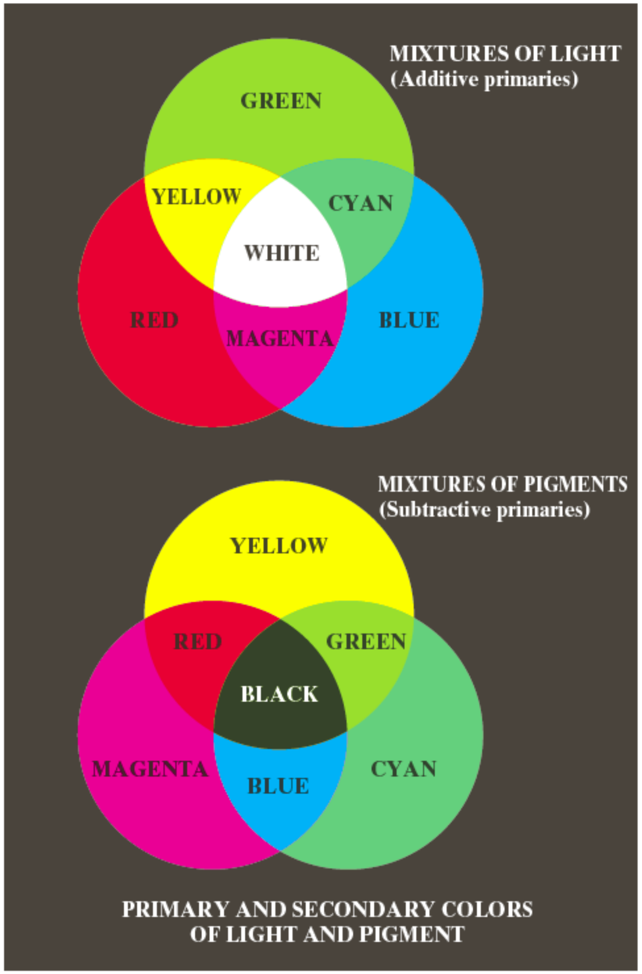
Color Illusions

Same color

Image from http://boingboing.net/2008/02/08/color-tile-optical-i.html

Primary & Secondary Colors

- Three primary color: red (R 65%), green (G 33%), and blue (B 2%).
- Secondary color: magenta (R + B), cyan (G+B), and yellow (R+G).



a
b

FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

Chromaticity

- Color perceptual attributes:
  - Brightness -- intensity (perceived Luminance)
  - Hue – perceived dominant color such as “redness”, “greenness”, …
  - Saturation – relative purity or the amount of white light mixed with a hue.
- Chromaticity: Hue + Saturation, independent of brightness

Chromaticity Coordinates

- The *chromaticity specifies the hue and saturation, but not* the brightness.

$$r = \frac{R}{R+G+B}, g = \frac{G}{R+G+B}, b = \frac{B}{R+G+B}.$$

r+g +b = 1 - Third component can always be computed from first two.

CIE Chromaticity Diagram

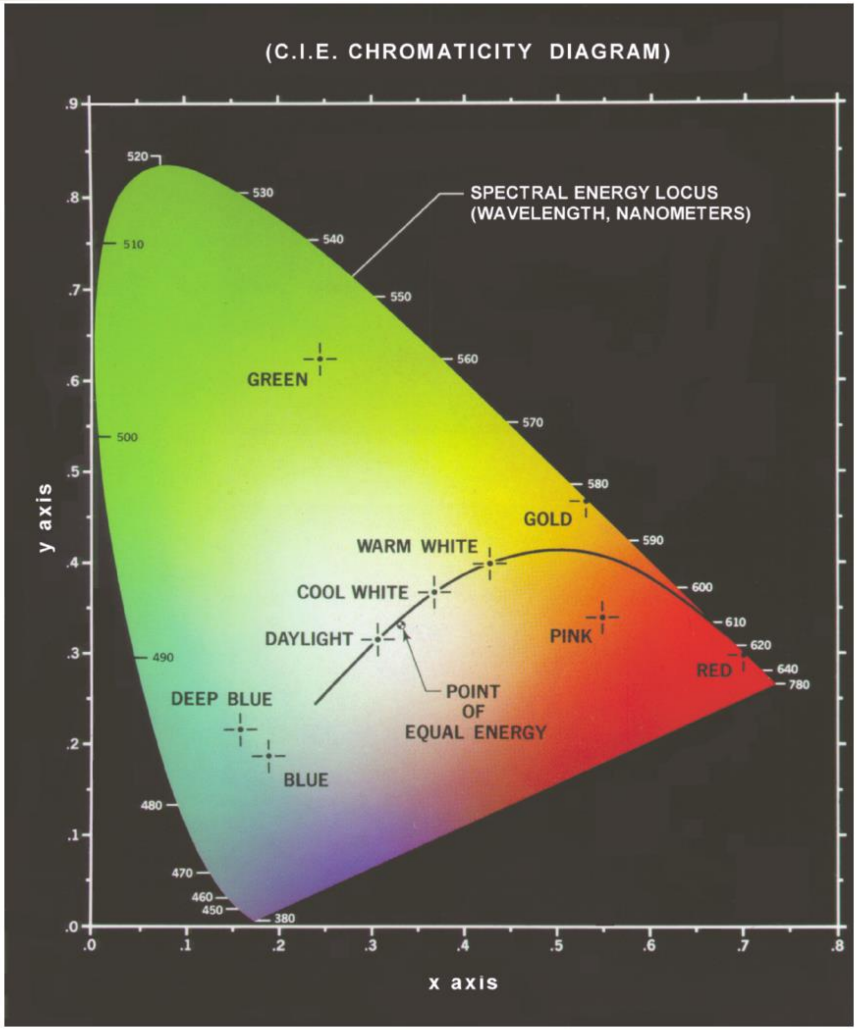


FIGURE 6.5 Chromaticity diagram. (Courtesy of the General Electric Co., Lamp Business Division.)

CIE: the International Commission on Illumination

Color Monitor & Printer Color Gamut

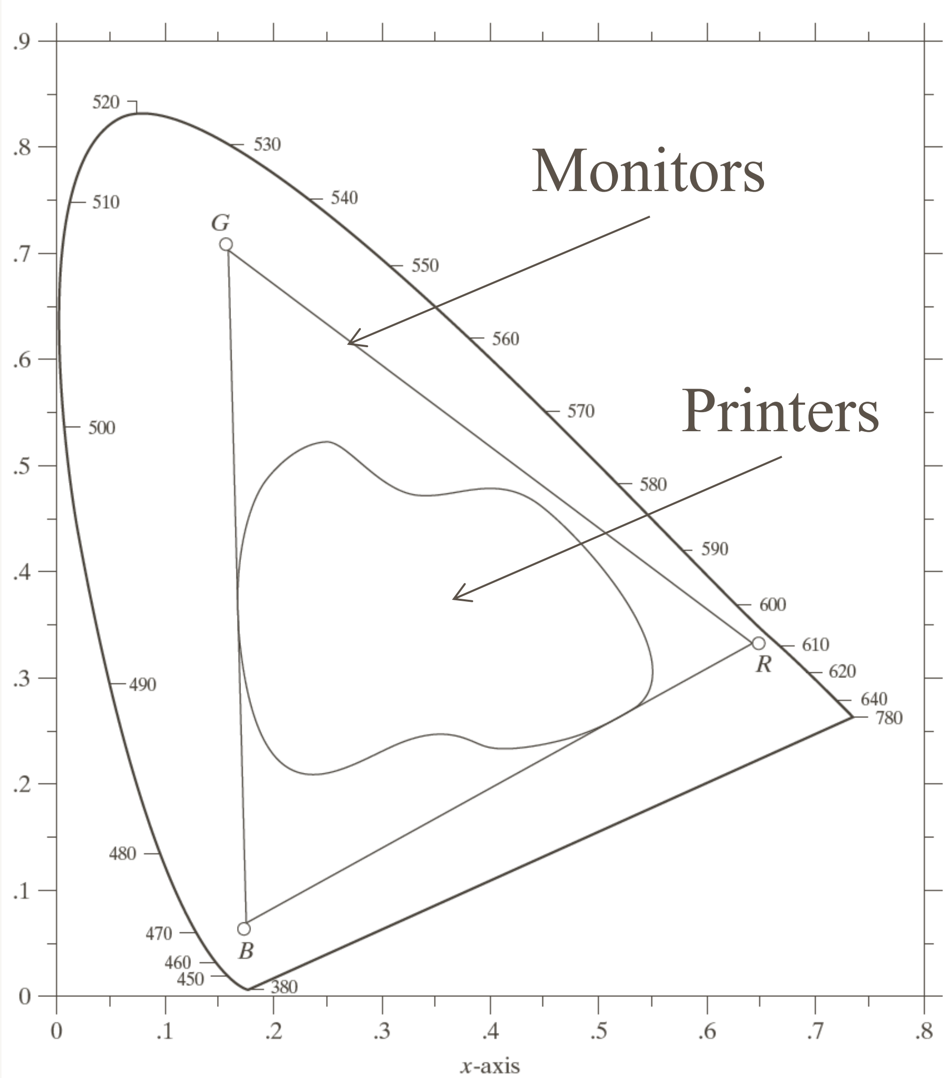
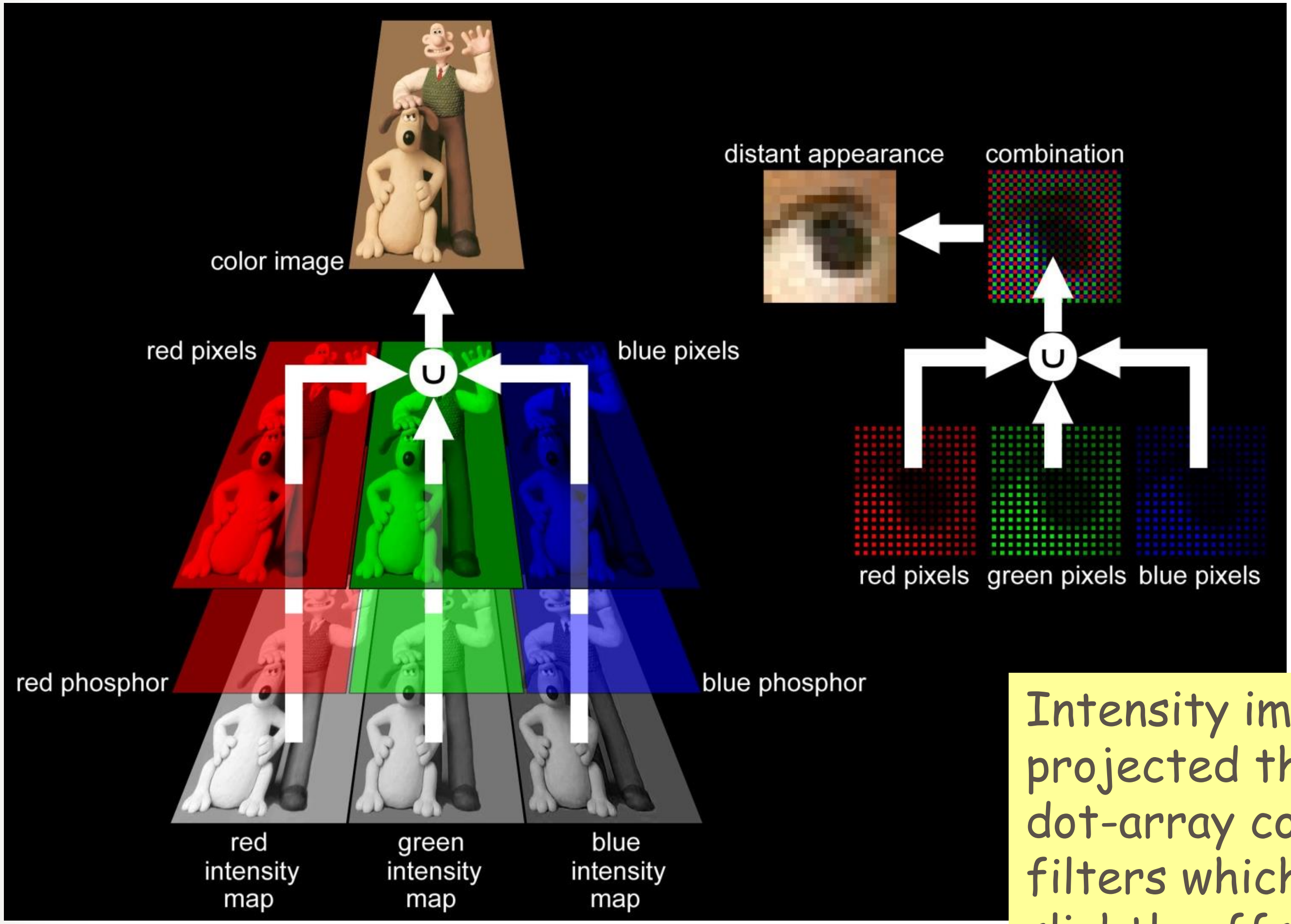


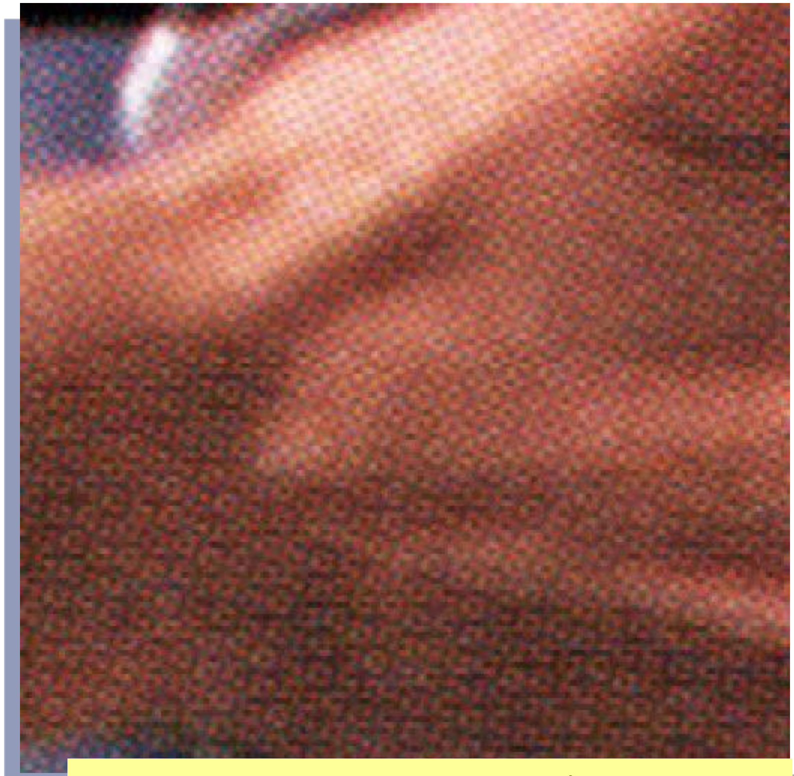
FIGURE 6.6
Typical color gamut of color monitors (triangle) and color printing devices (irregular region).

Color Images on a CRT or LCD Display



Intensity images are projected through dot-array color filters which are slightly offset from one another.

Color Images In Print



Cyan (C)
Magenta (M)
Yellow (Y)
Key (K), -- Black

Images are separated into four color bands, each of which is printed as a grid regularly spaced dots. A dot's diameter varies in proportion to the intensity of the color.

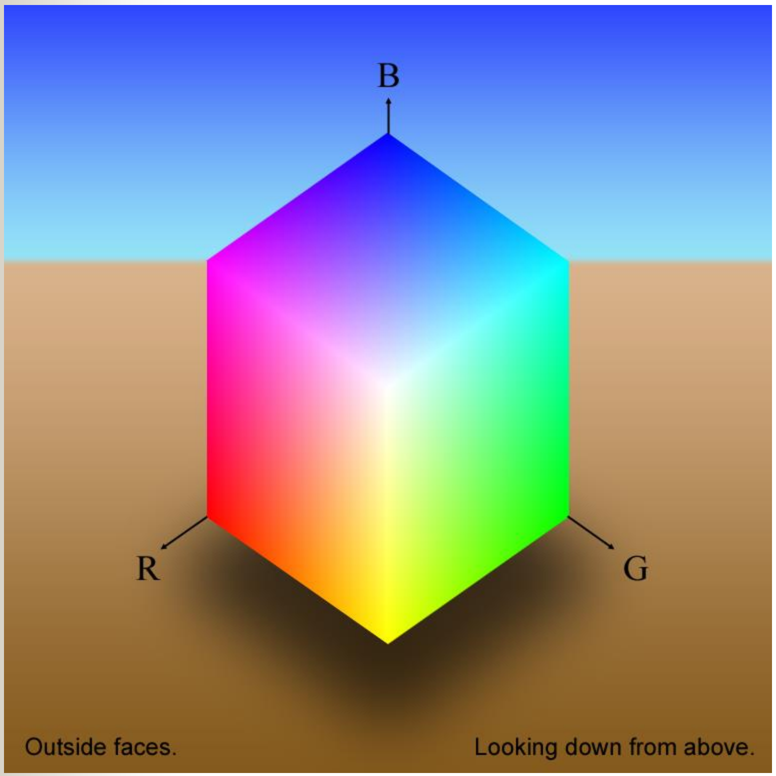
Color Images In Print

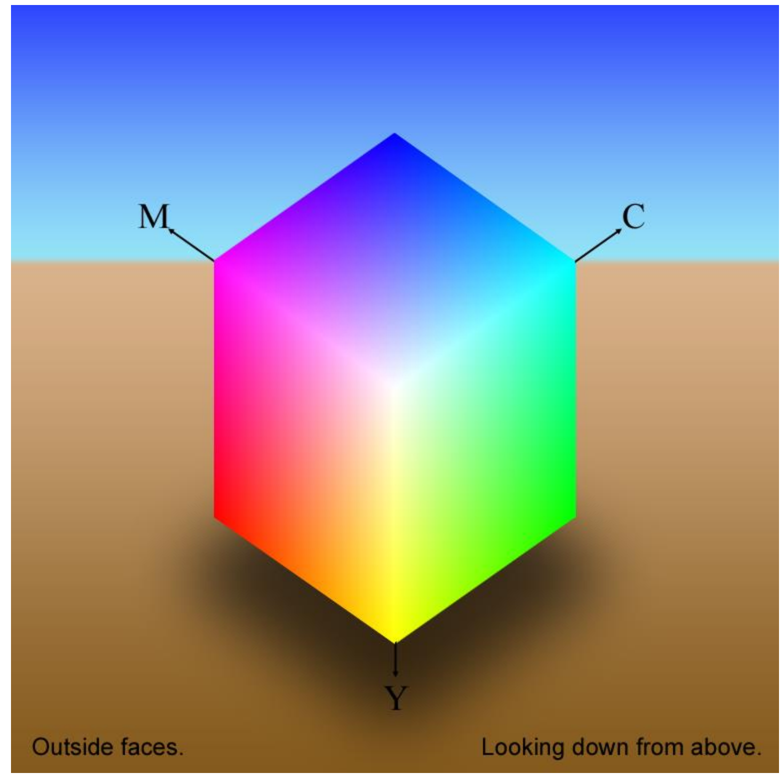
The four colors are magenta, cyan, yellow, and black (CMYK color model)

Color Models

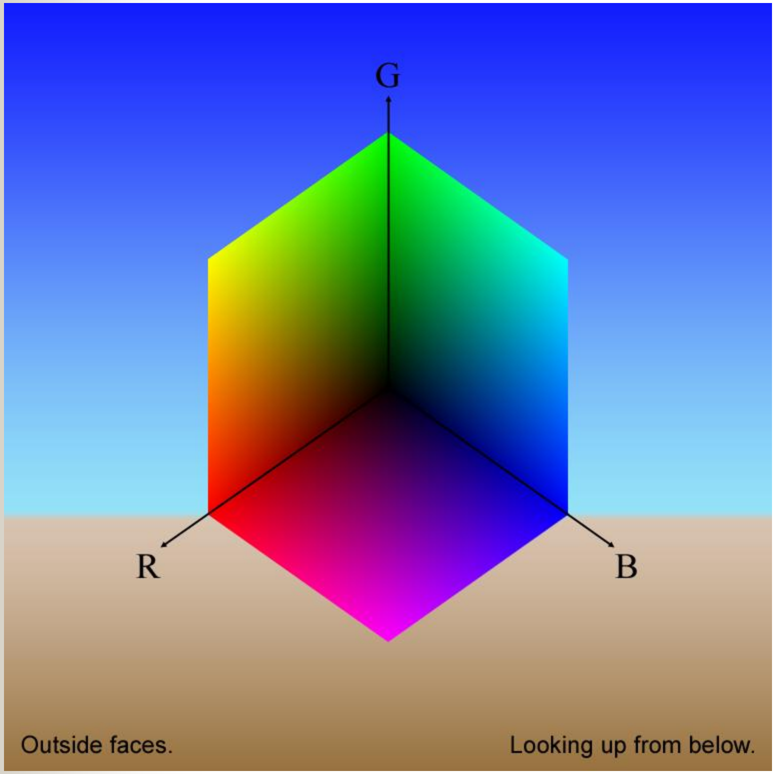
- RGB
- CMY
- CMYK
- HSI

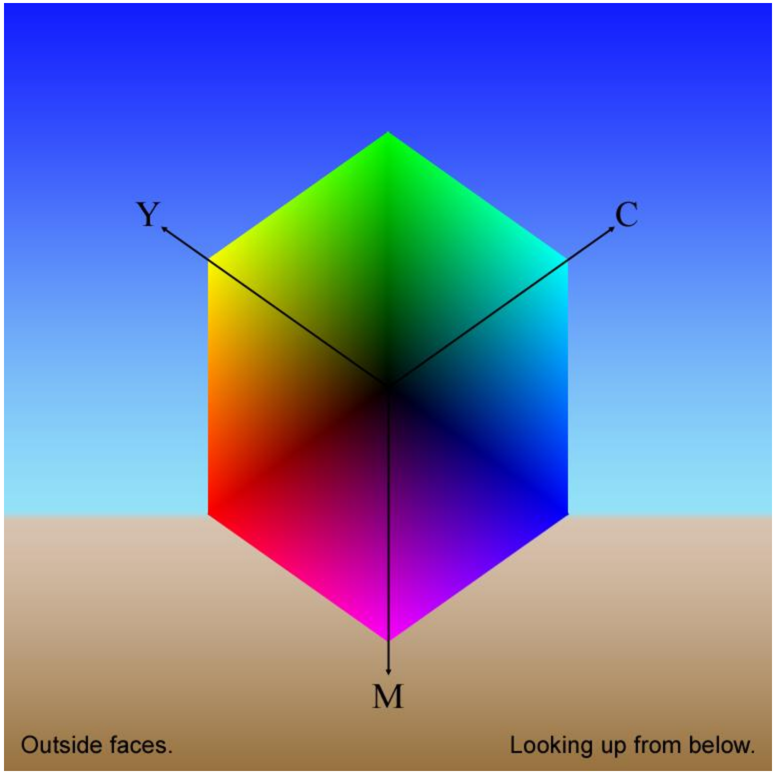
RBG Color Cube (Outer)



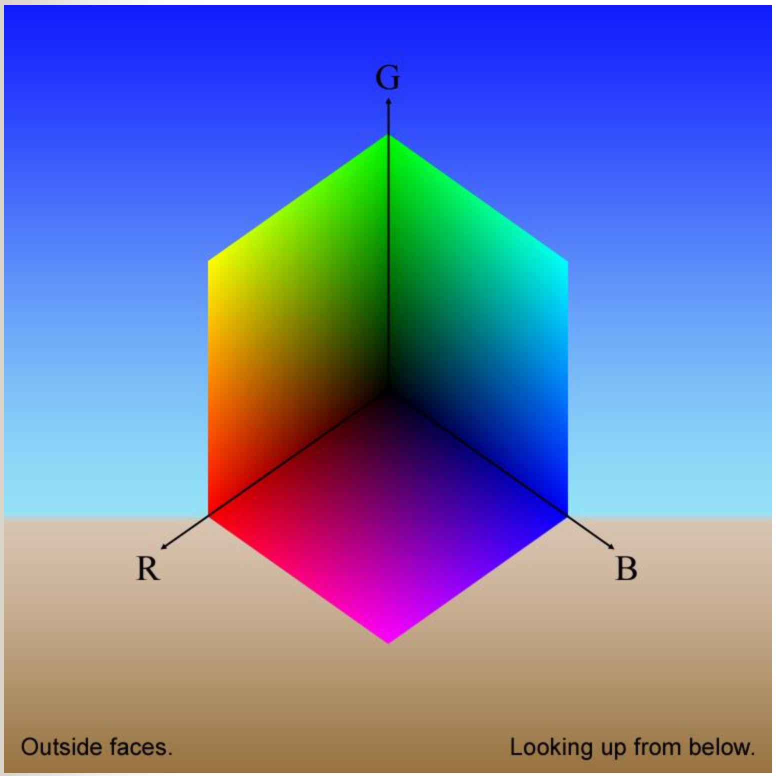


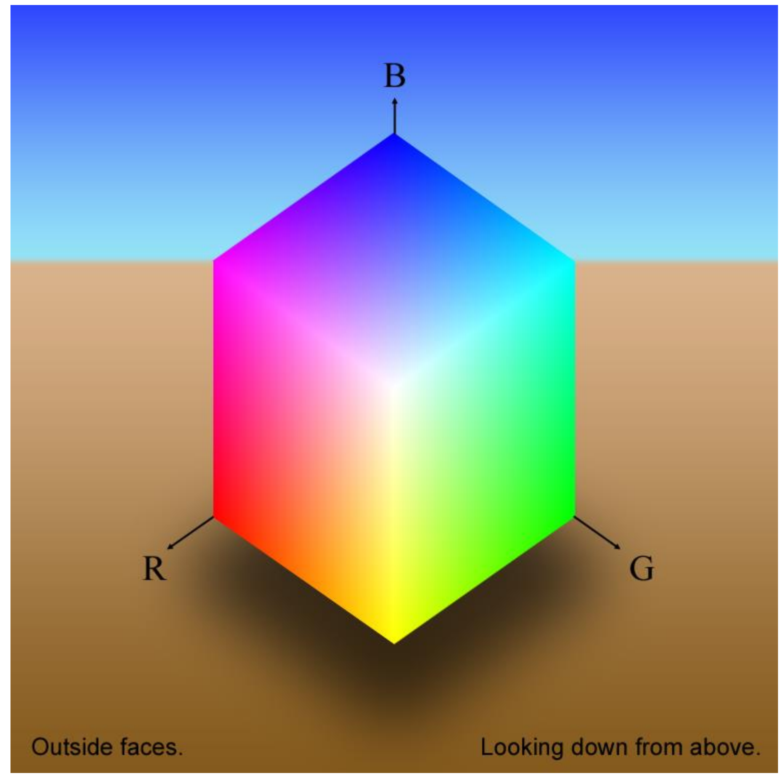
RBG Color Cube (inner)



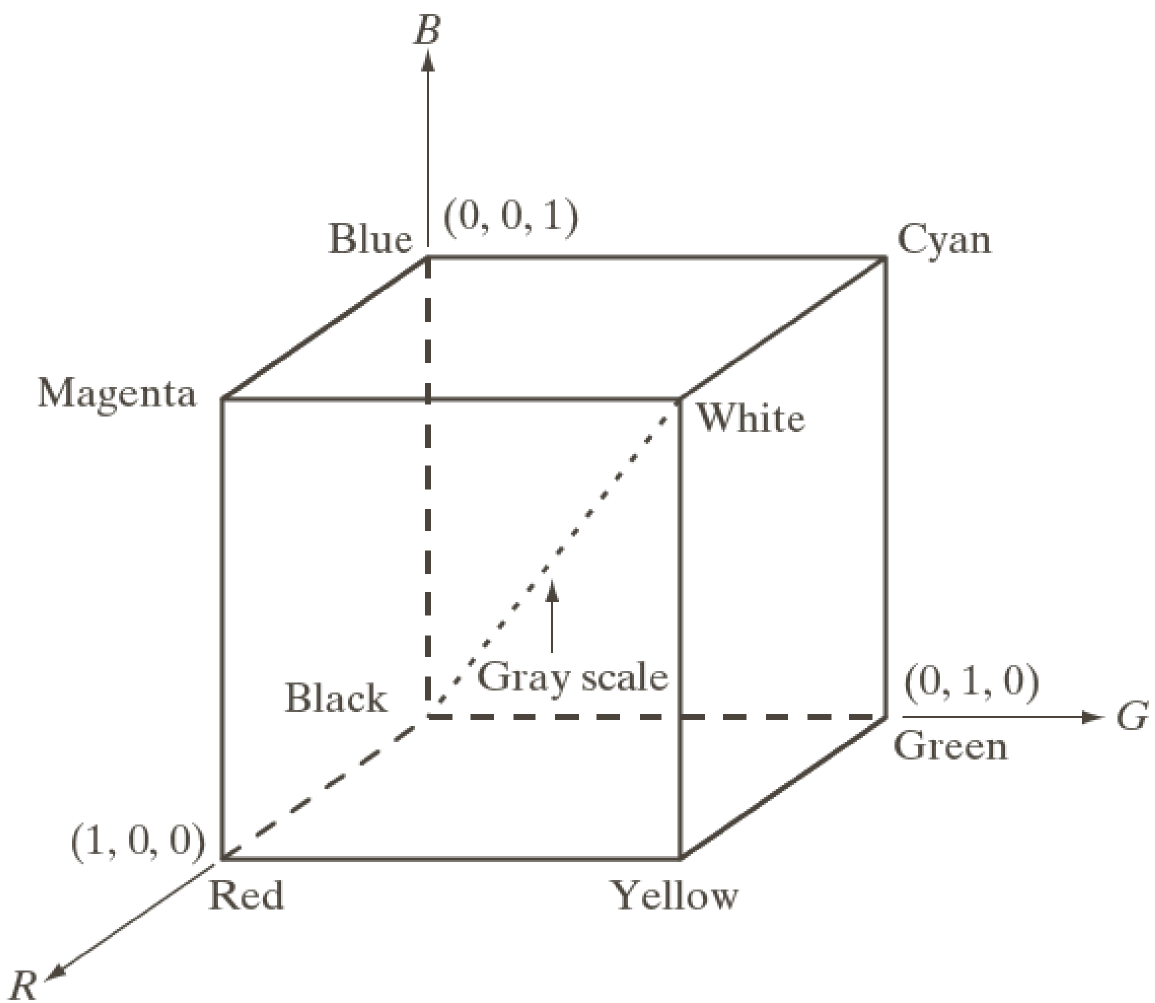


RBG Color Cube (inner and outer)





RBG Color Cube



RGB to CMY Color Model

C– cyan; M—magenta; Y—yellow
CMYK– add "black" color

$C = 1 - R$
$M = 1 - G$
$Y = 1 - B$

RGB – YIQ Color Model

color primary system adopted by NTSC for color television broadcasting

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

RGB – YUV Color Model

Similar to YIQ but adopted by PAL for color television broadcasting

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.437 \\ 0.615 & -0.515 & -0.100 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

L*a*b* color model

$$L^* = 116\, f(Y/Y_n) - 16$$

$$a^* = 500\,[f(X/X_n) - f(Y/Y_n)]$$

$$b^* = 200\,[f(Y/Y_n) - f(Z/Z_n)]$$

$$f(t) = \begin{cases} t^{1/3} & t > (6/29)^3 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{otherwise} \end{cases}$$

HSI Color Model

- H – hue, S-saturation, I- intensity.

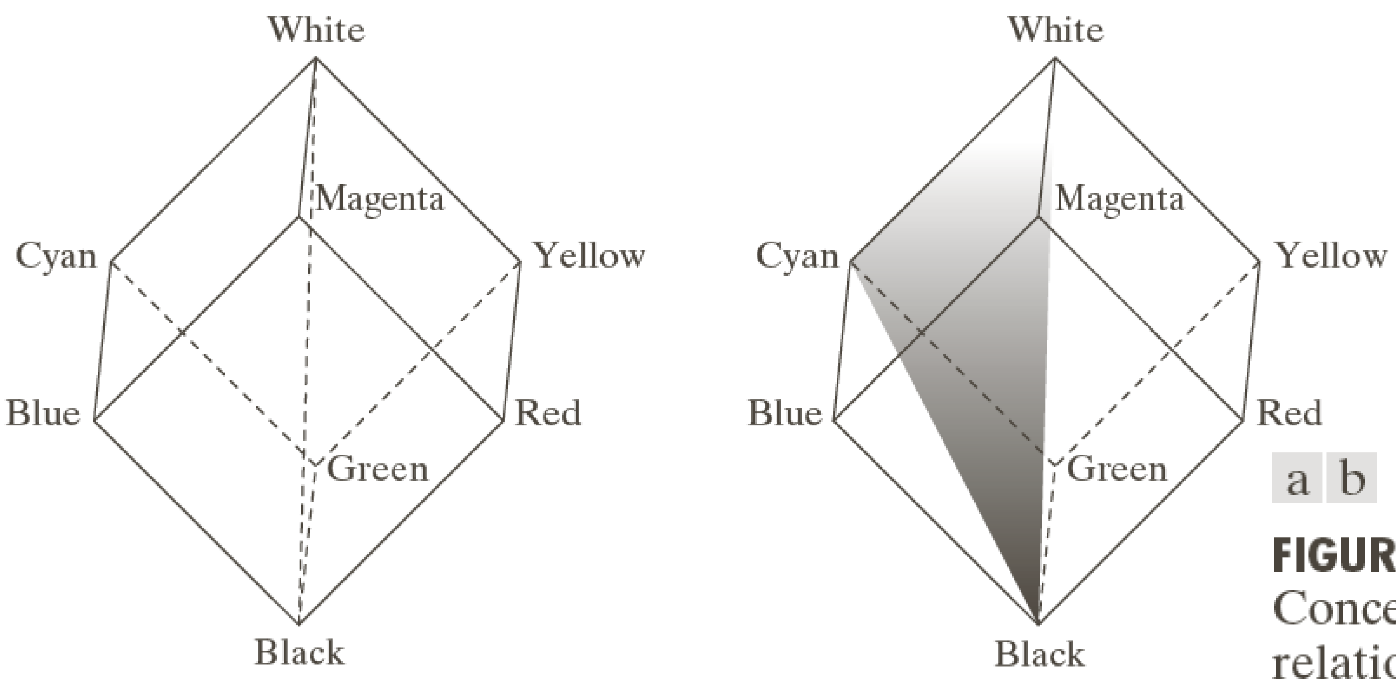
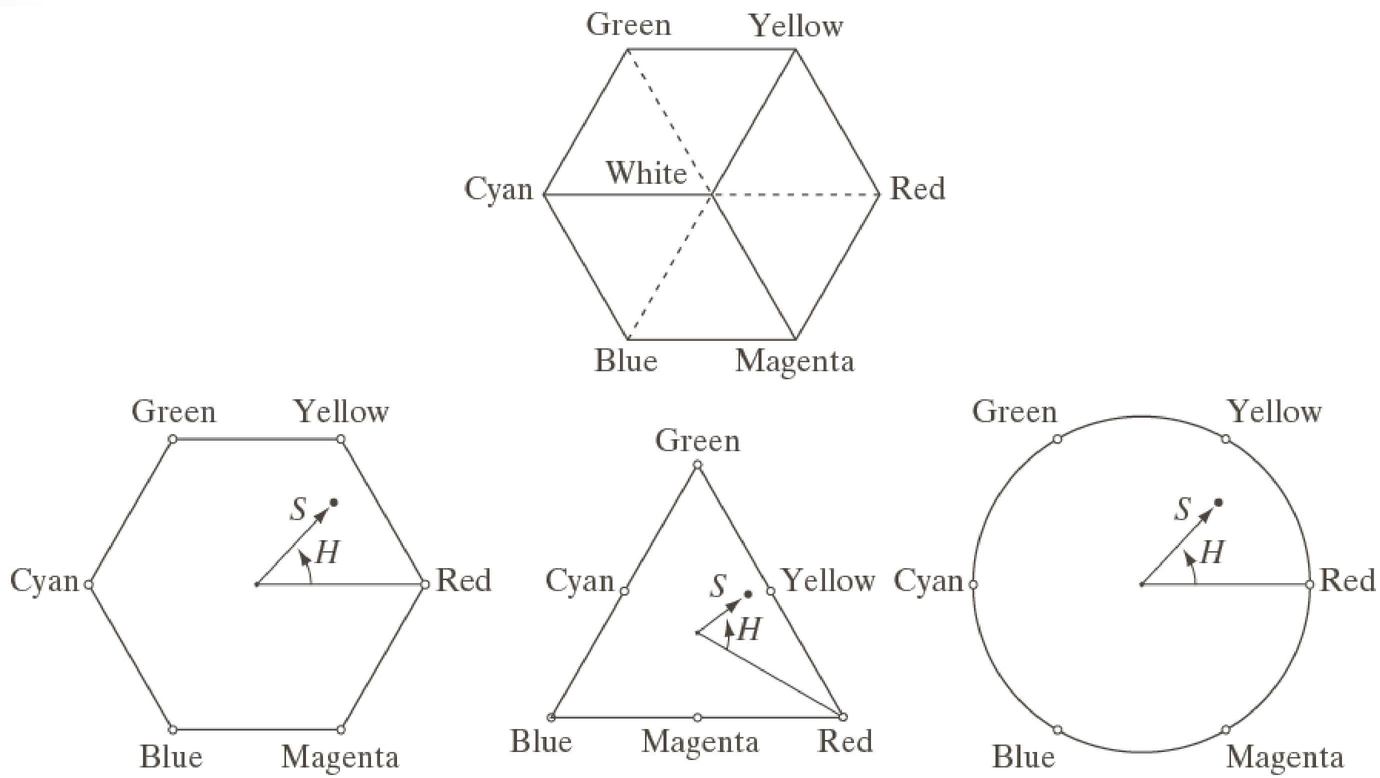


FIGURE 6.12 Conceptual relationships between the RGB and HSI color models.

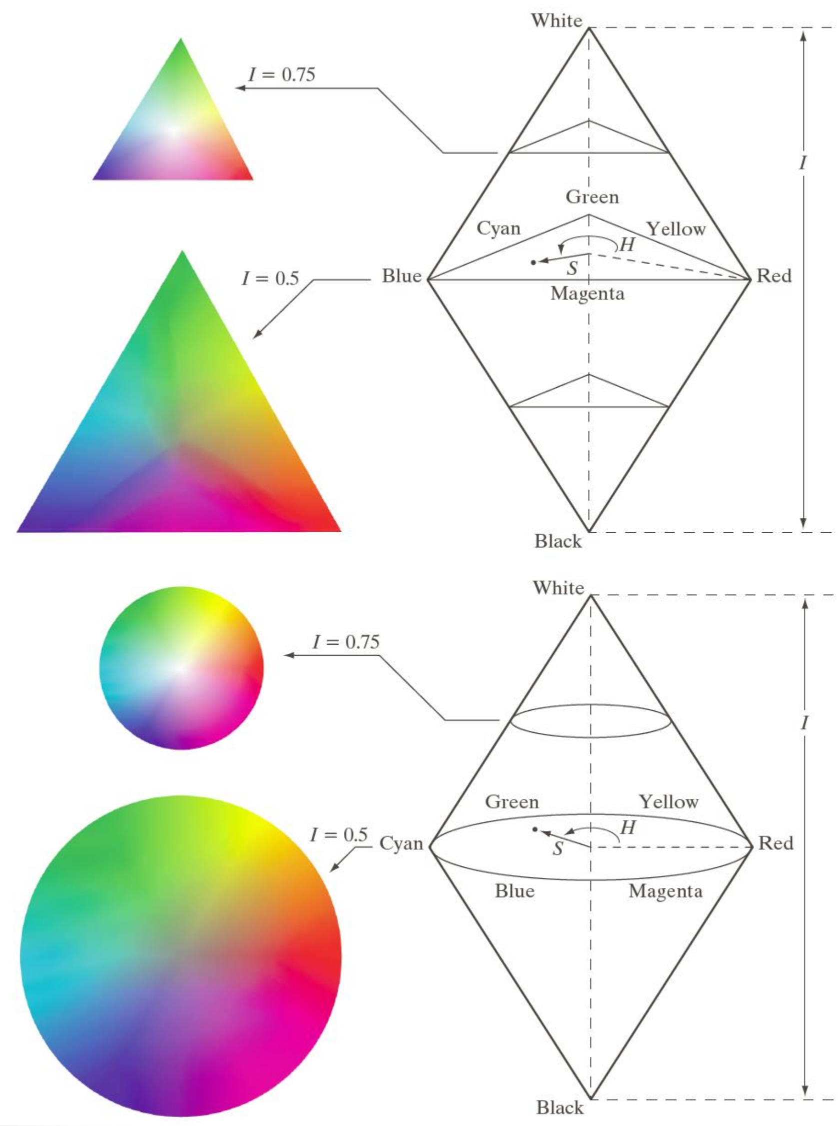
HSI Color Model



a
b c d

FIGURE 6.13 Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

HSI Color Model



a
b

FIGURE 6.14 The HSI color model based on (a) triangular and (b) circular color planes. The triangles and circles are perpendicular to the vertical intensity axis.

RGB to HSI Color Model

$$r = \frac{R}{R+G+B},\ g = \frac{G}{R+G+B},\ b = \frac{B}{R+G+B}.$$

$$h = \cos^{-1}\left\{\frac{0.5\cdot[(r-g)+(r-b)]}{\left[(r-g)^2+(r-b)(g-b)\right]^{1/2}}\right\} \qquad h\in[0,\pi] \text{ for } b\le g$$

$$h = 2\pi-\cos^{-1}\left\{\frac{0.5\cdot[(r-g)+(r-b)]}{\left[(r-g)^2+(r-b)(g-b)\right]^{1/2}}\right\} \qquad h\in[\pi,2\pi] \text{ for } b> g$$

$$s = 1-3\cdot\min(r,g,b);\qquad s\in[0,1]$$

$$i = (R+G+B)/(3\cdot 255);\qquad i\in[0,1].$$

RGB to HSI Color Model

For convenience, h, s and i values are converted in the ranges of [0,360], [0,100], [0, 255], respectively , by:

$$H = h \times 180 / \pi; \quad S = s \times 100 \text{ and } \quad I = i \times 255.$$

HSI to RGB Color Model

$h = H \cdot \pi / 180$; $s = S / 100$; $i = I / 255$

$$x = i \cdot (1 - s)$$

$$y = i \cdot \left[1 + \frac{s \cdot \cos(h)}{\cos(\pi / 3 - h)}\right]$$

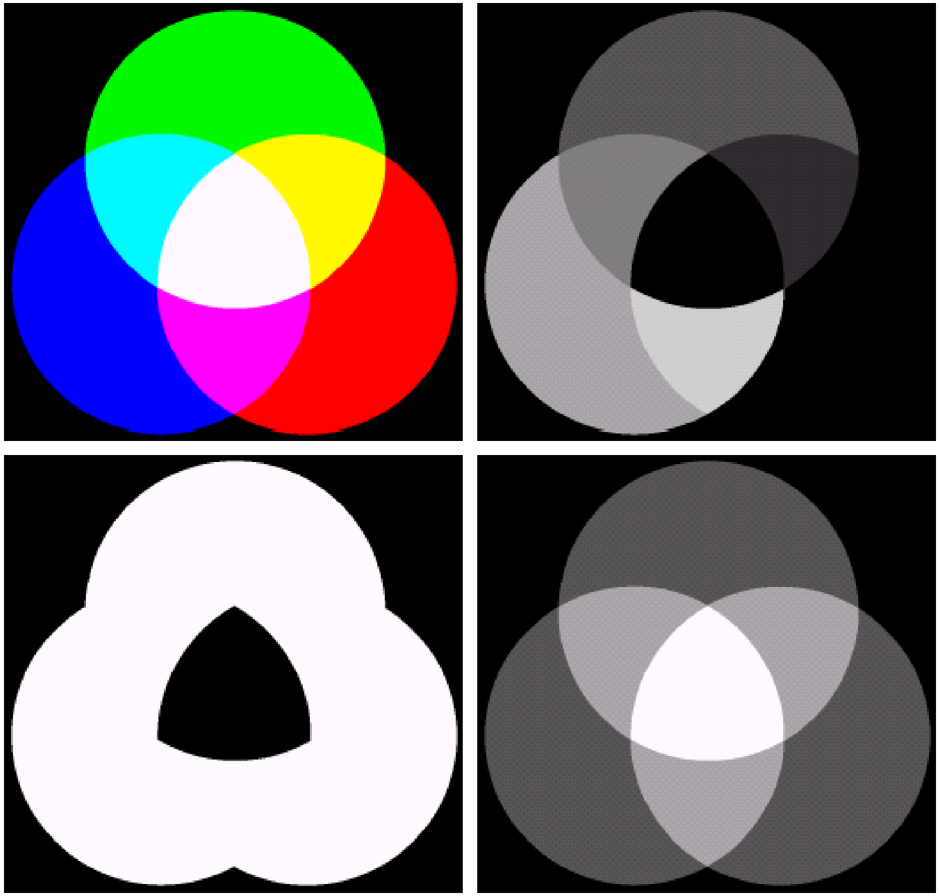
$z = 3i - (x + y)$;

when $h < 2\pi / 3$, $b = x$; $r = y$ and $g = z$.

when $2\pi / 3 \leq h < 4\pi / 3$, $h = h - 2\pi / 3$, and $r = x$; $g = y$ and $b = z$.

when $4\pi / 3 \leq h < 2\pi$, $h = h - 4\pi / 3$, and $g = x$; $b = y$ and $r = z$.

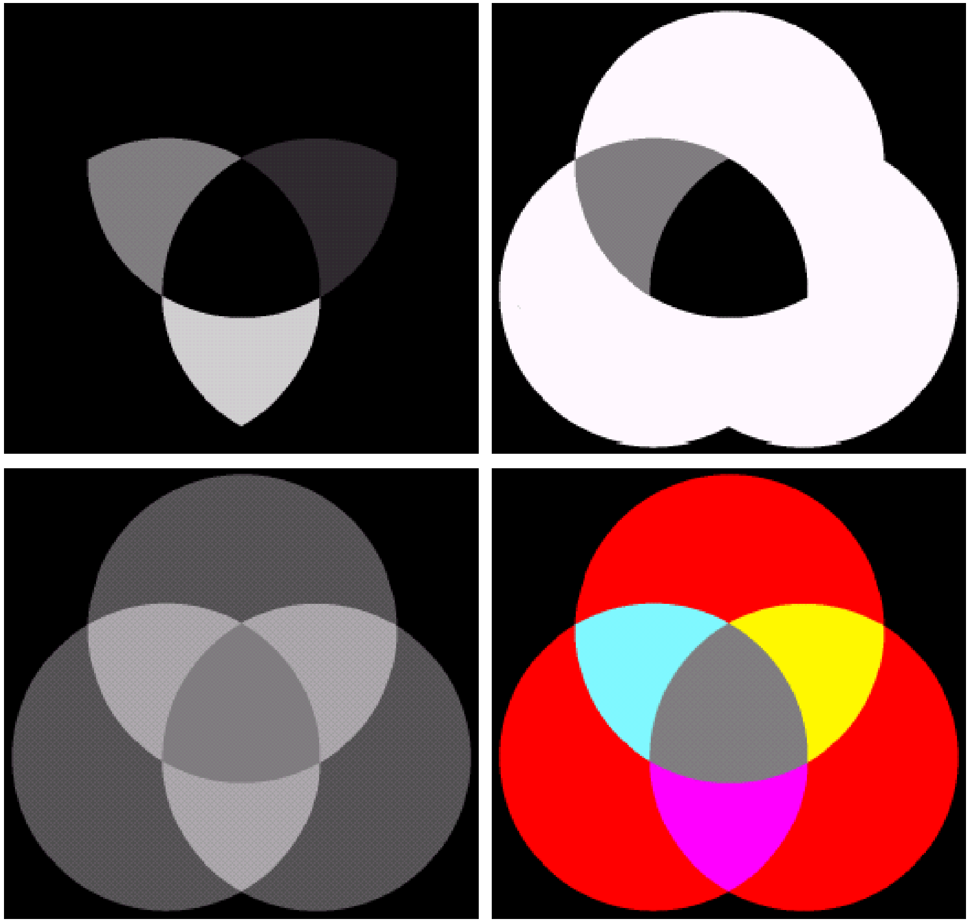
HSI example



a b
c d

FIGURE 6.16 (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.

HSI example

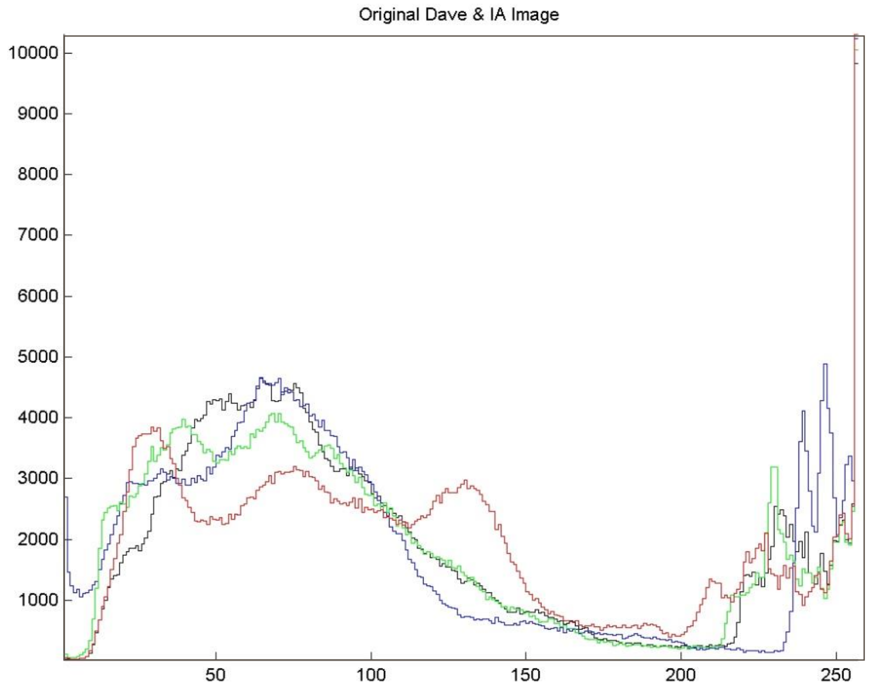


a b
c d

FIGURE 6.17 (a)–(c) Modified HSI component images. (d) Resulting RGB image. (See Fig. 6.16 for the original HSI images.)

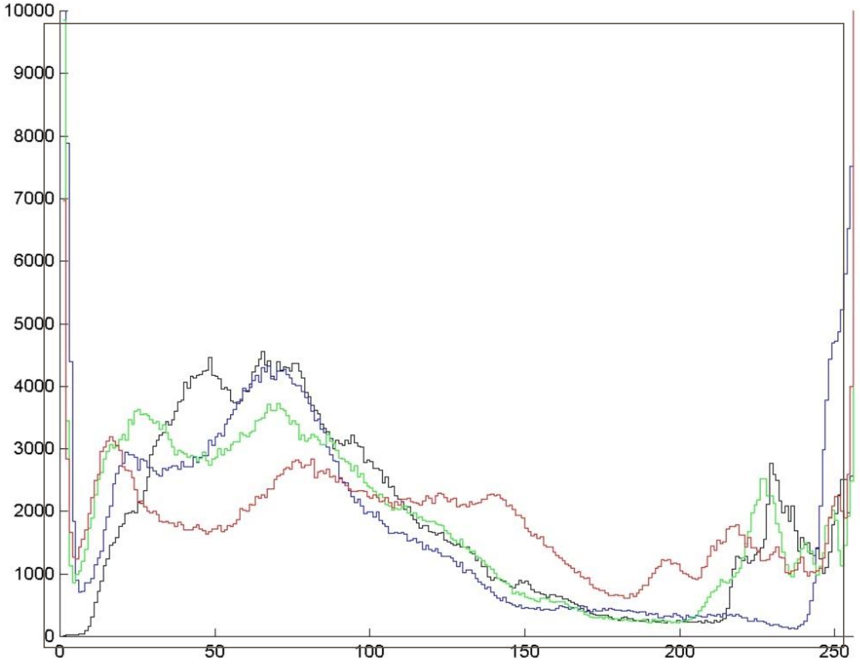
Saturation Adjustment

original



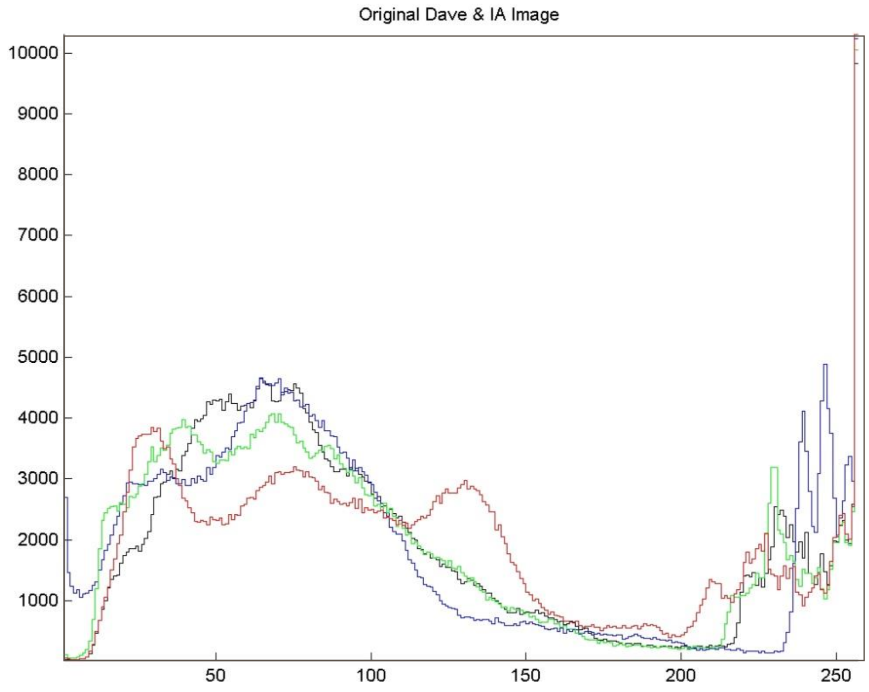
Saturation Adjustment

saturation + 50%



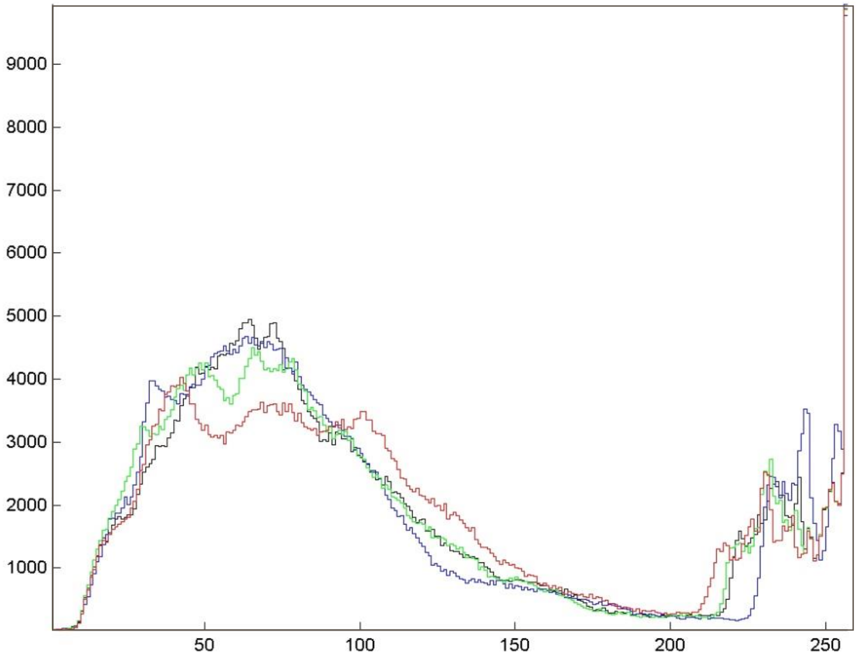
Saturation Adjustment

original



Saturation Adjustment

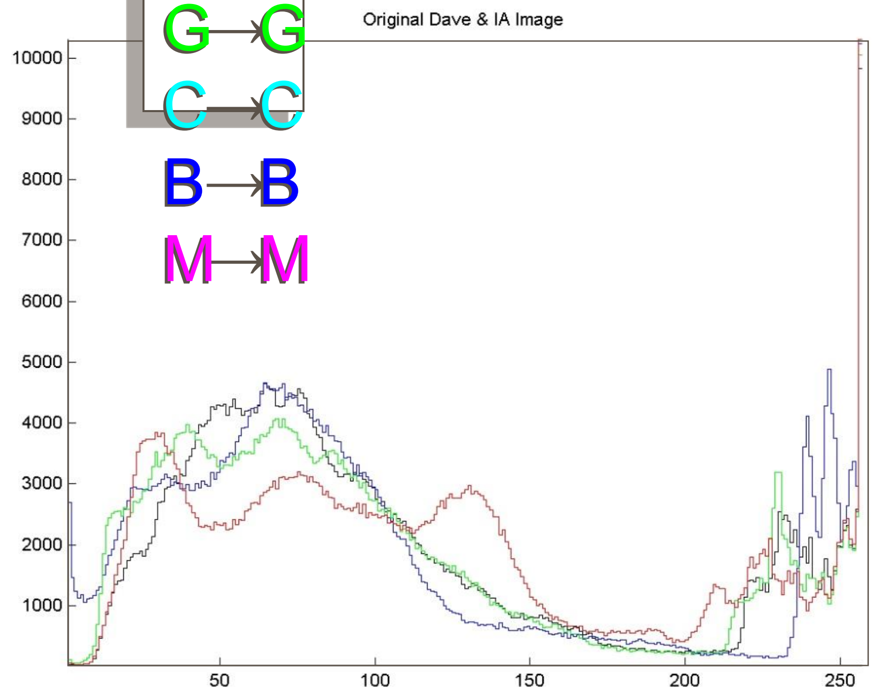
saturation - 50%



Hue Shifting

R→R
Y→Y
G→G
C→C
B→B
M→M

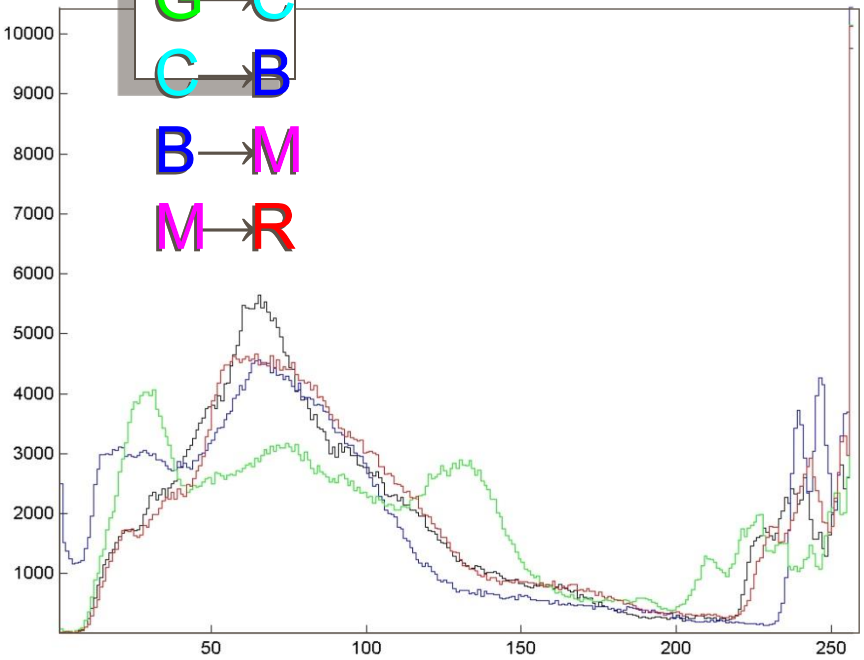
original



Hue Shifting

R→Y
Y→G
G→C
C→B
B→M
M→R

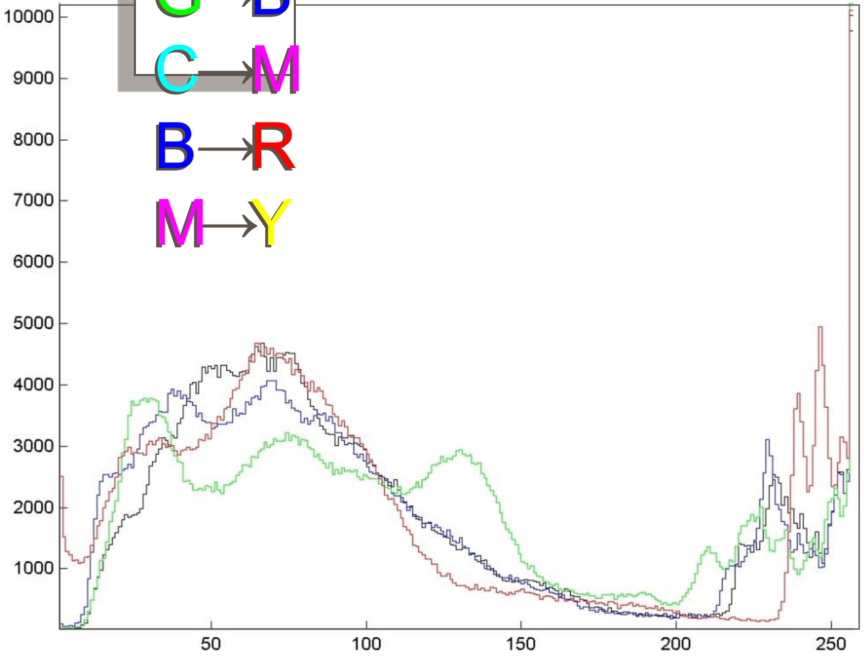
hue + 60°



Hue Shifting

R→G
Y→C
G→B
C→M
B→R
M→Y

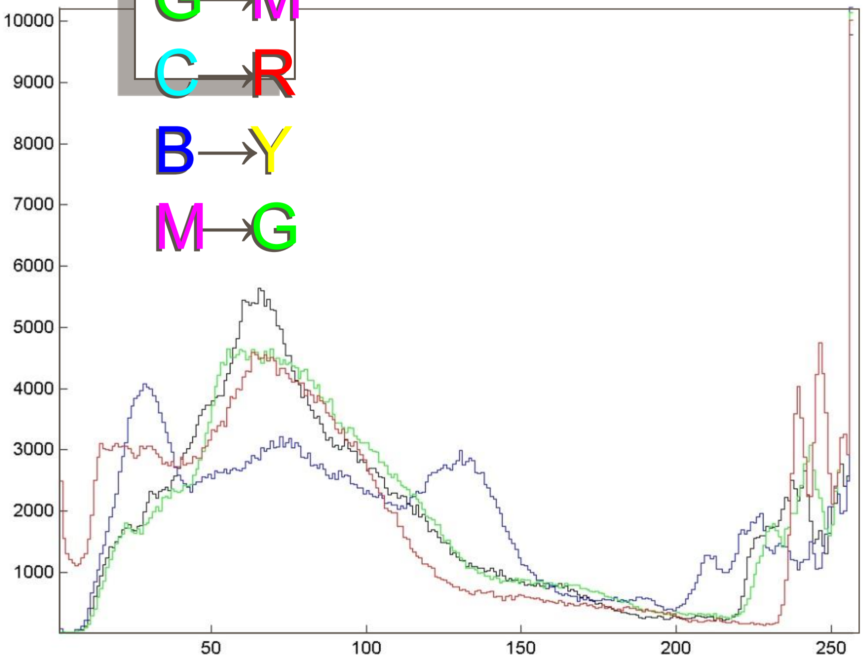
hue + 120°



Hue Shifting

R→C
Y→B
G→M
C→R
B→Y
M→G

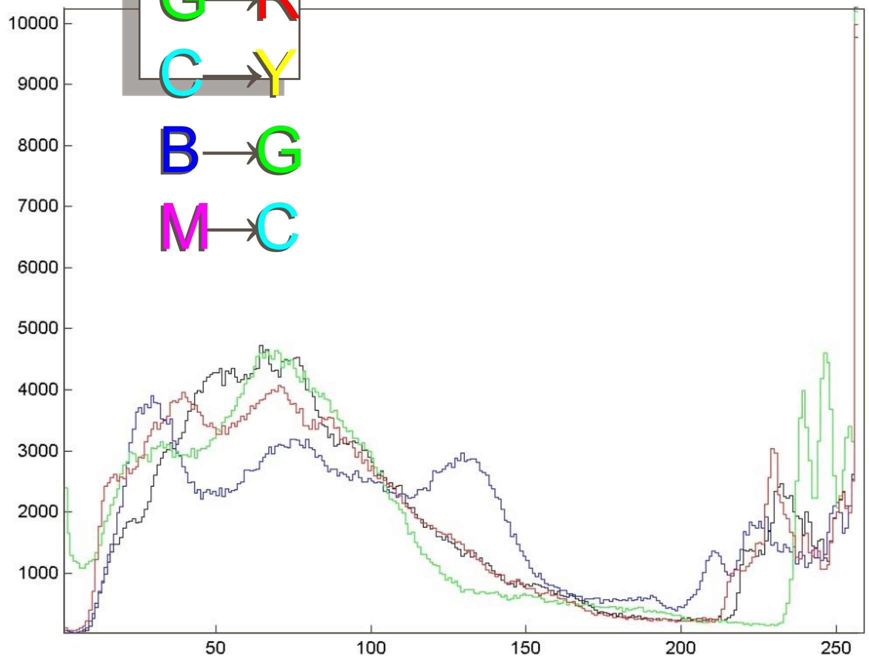
hue + 180°



Hue Shifting

hue + 240°

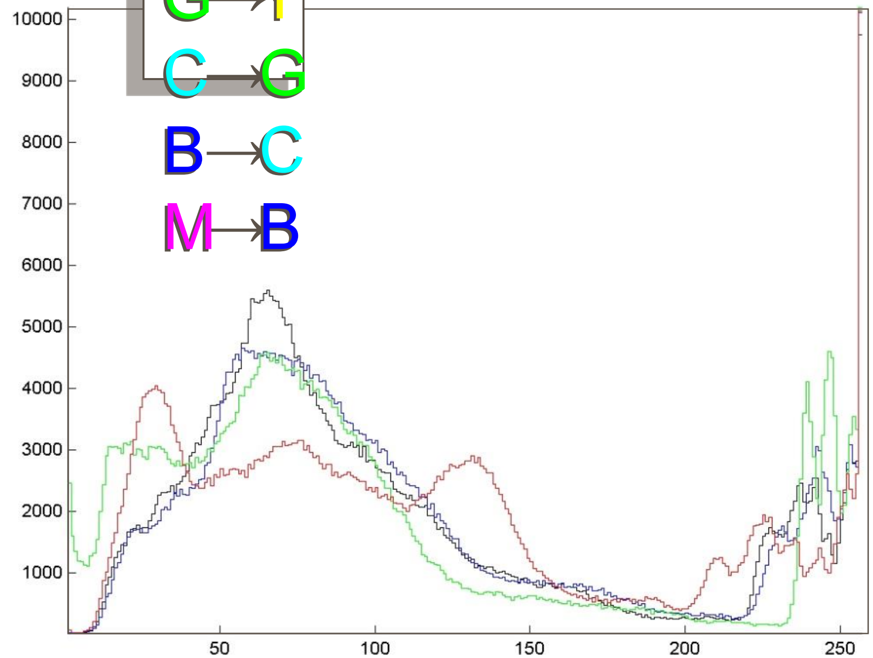
R→B
Y→M
G→R
C→Y
B→G
M→C



Hue Shifting

hue + 300°

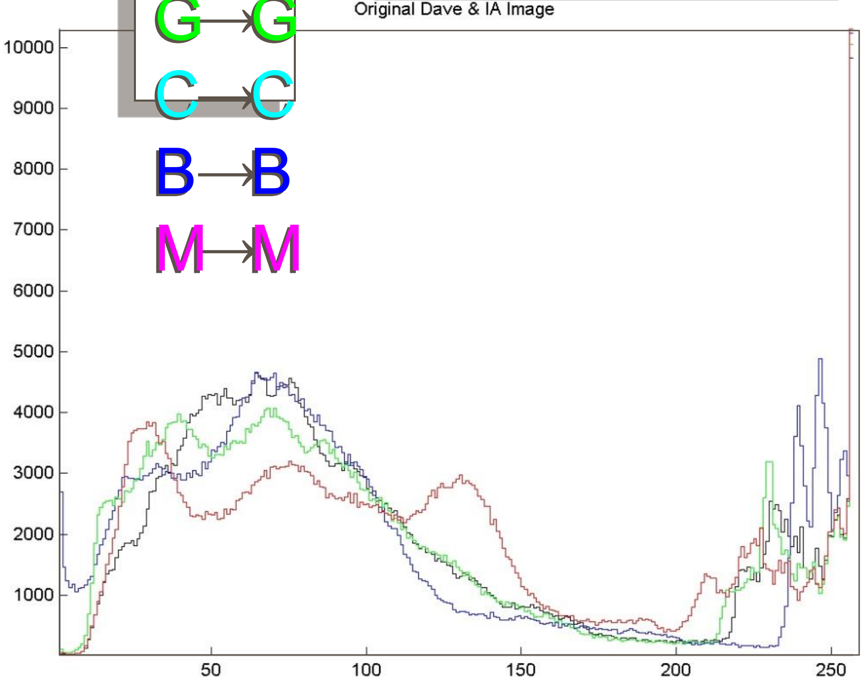
R→M
Y→R
G→Y
C→G
B→C
M→B



Hue Shifting

R→R
Y→Y
G→G
C→C
B→B
M→M

hue + 360° = original



Pseudocolor Image Processing

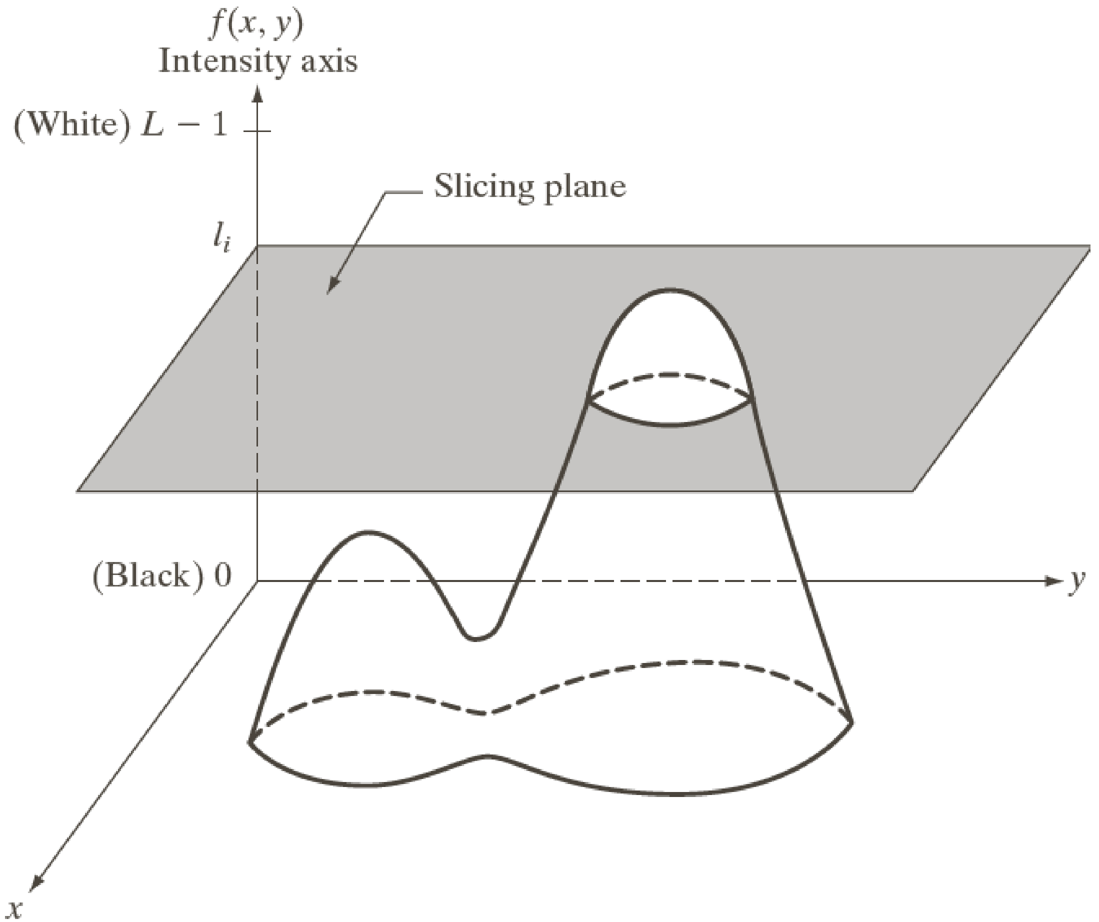
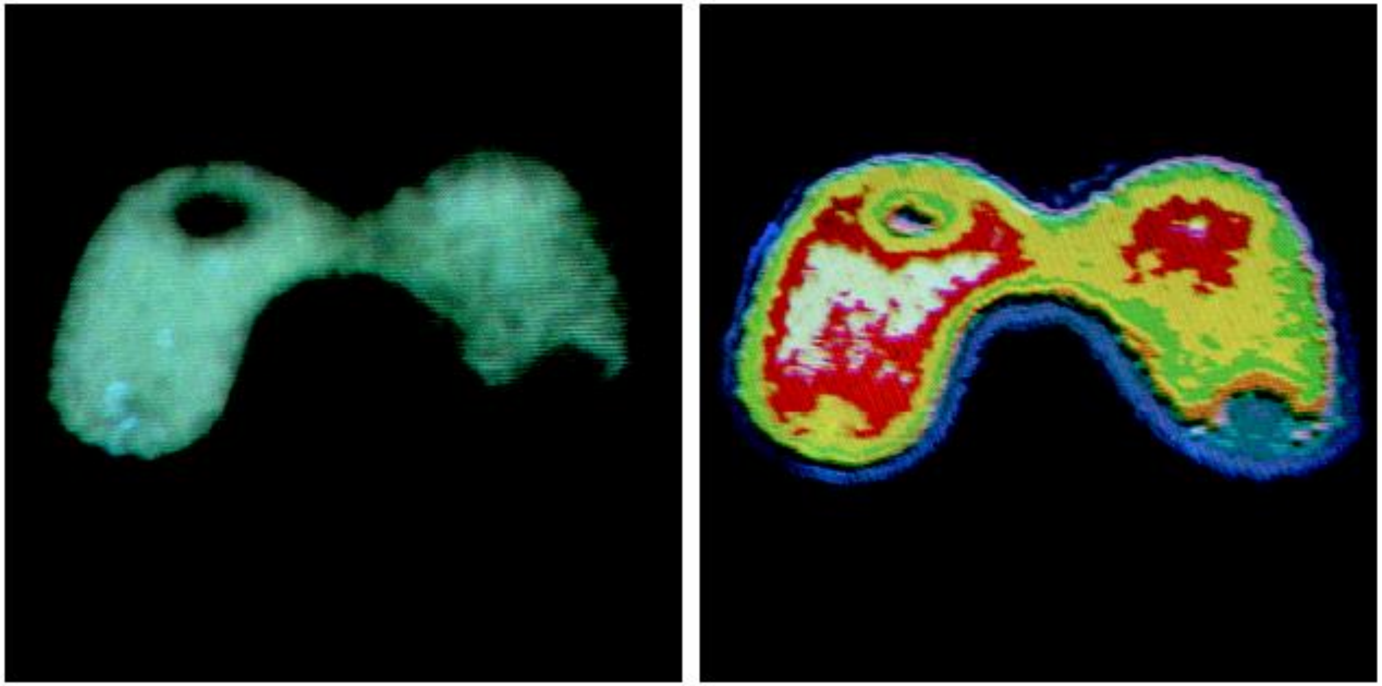


FIGURE 6.18 Geometric interpretation of the intensity-slicing technique.

Pseudocolor Image example



a b

FIGURE 6.20 (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

Intensity to Color

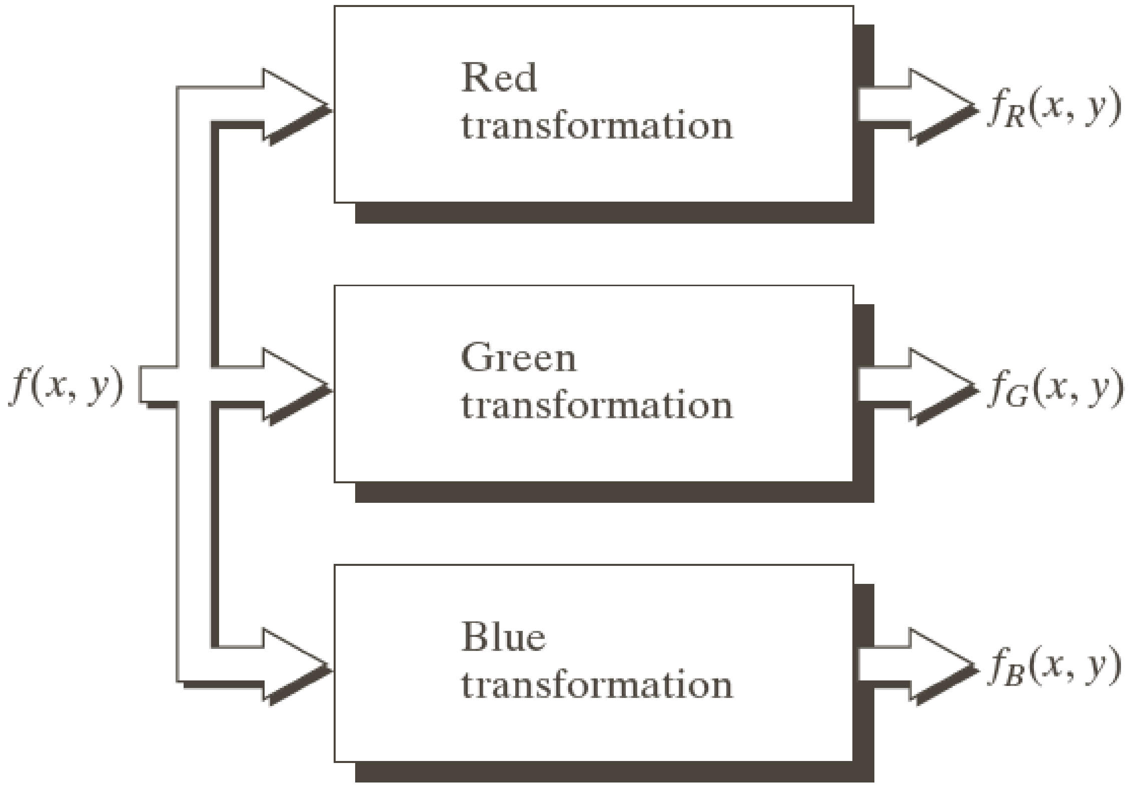
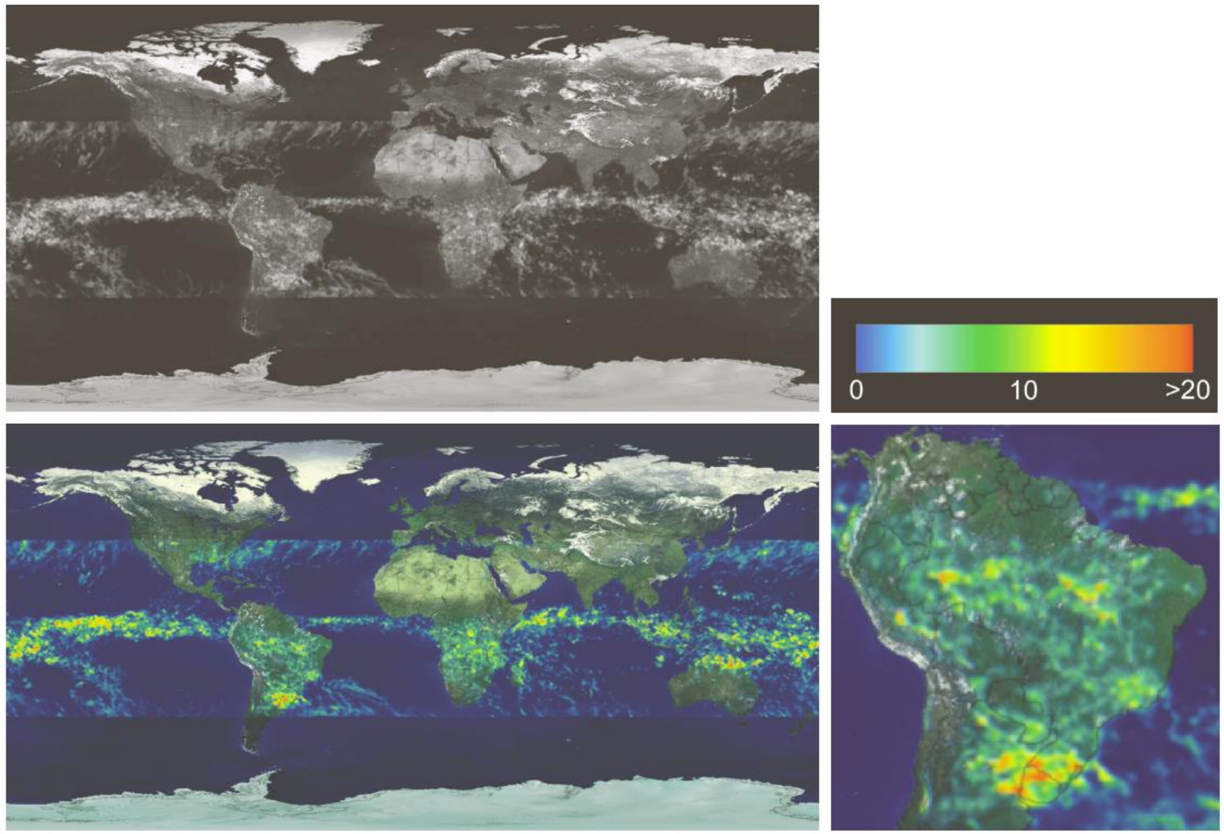


FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

Pseudocolor Image example

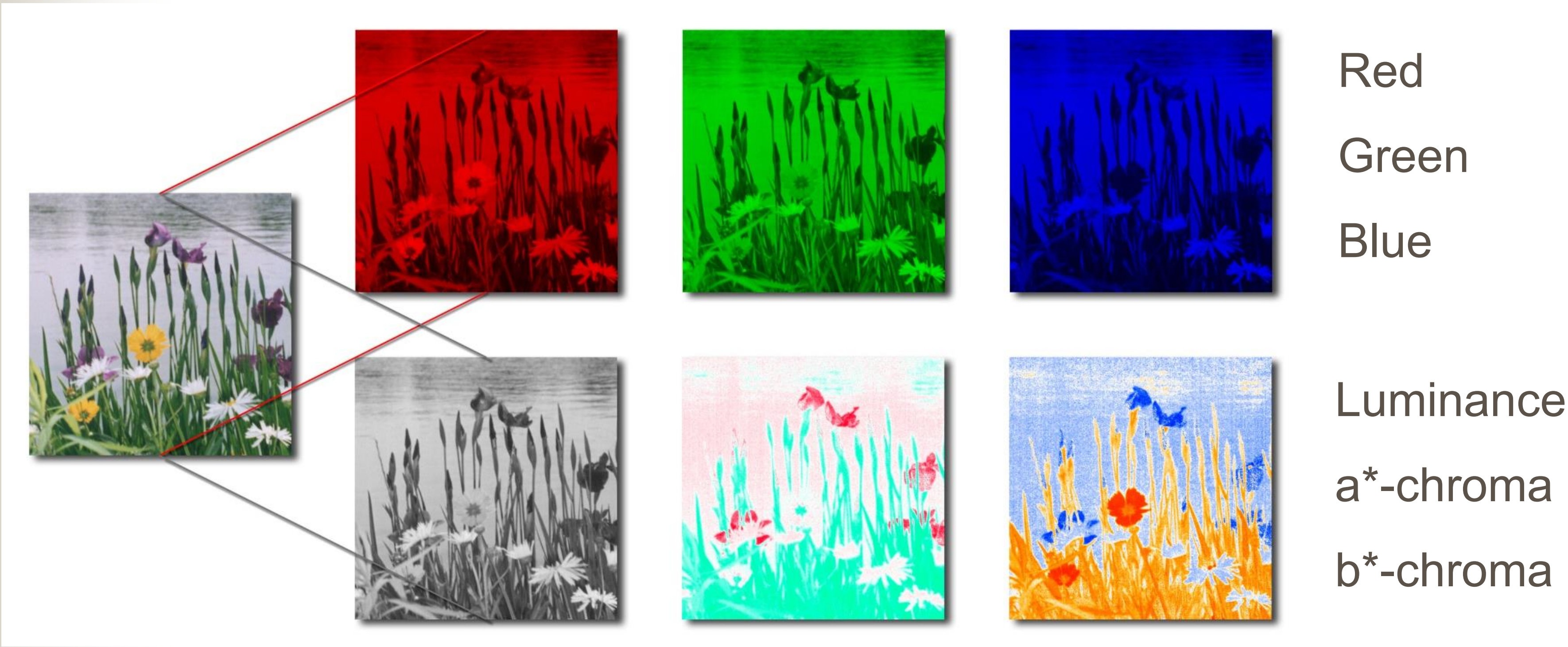


a b
c d

FIGURE 6.22 (a) Gray-scale image in which intensity (in the lighter horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)

Color Image

are represented by three bands (not uniquely) *e.g.*, R, G, & B or L, a*, & b*.



Color Image Processing

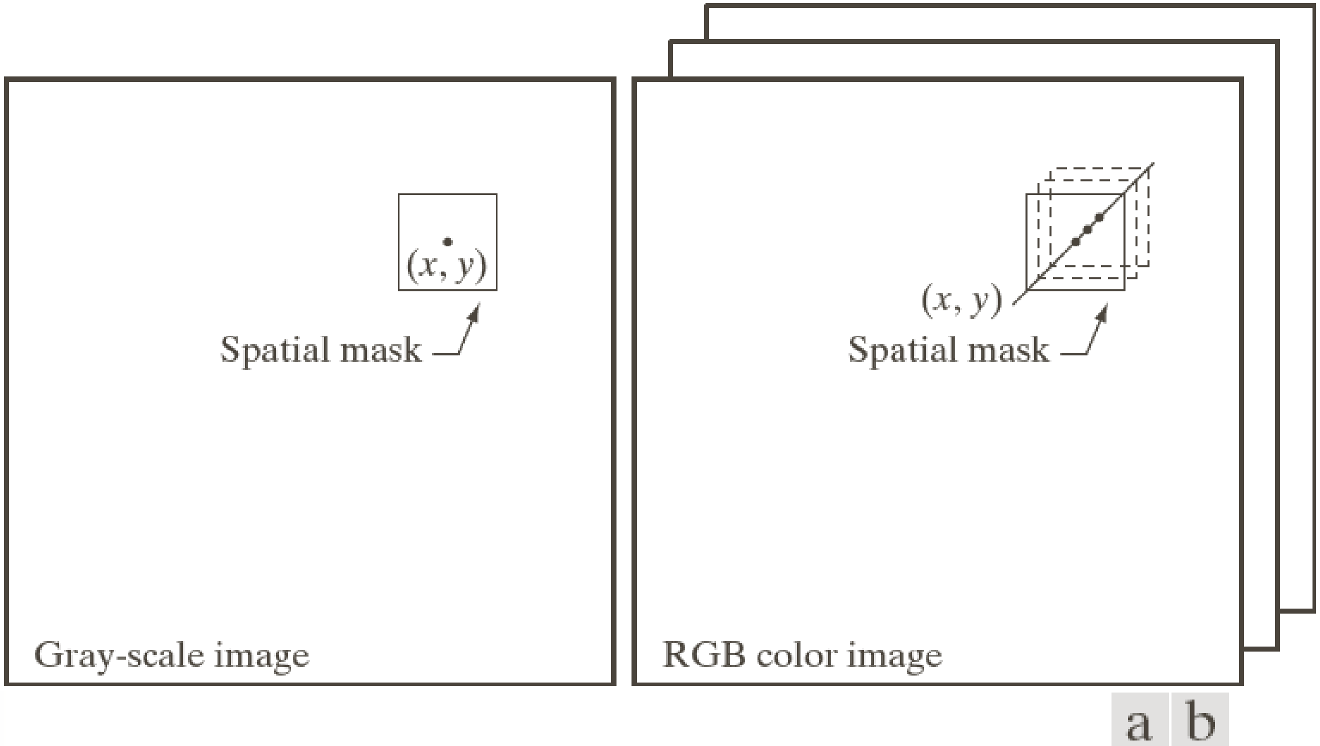


FIGURE 6.29
Spatial masks for gray-scale and RGB color images.

Color Correction

Global changes in the coloration of an image to alter its tint, its hues or the saturation of its colors with minimal changes to its luminant features

Color Transformation

- is a point process; the transformation is applied to each pixel as a function of its color alone.
- Each pixel is vector valued, therefore the transformation is a vector space operator.

Color Vector Space Operators

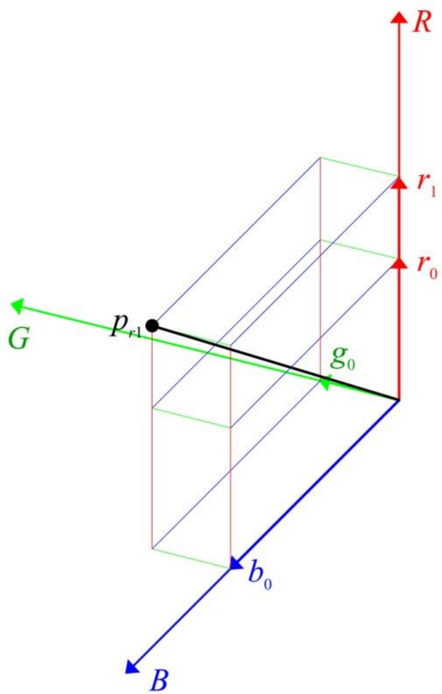
Linear operators are matrix multiplications

$$\begin{bmatrix} r_1 \\ g_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} r_0 \\ g_0 \\ b_0 \end{bmatrix}$$

$$\begin{bmatrix} r_1 \\ g_1 \\ b_1 \end{bmatrix} = 255 \cdot \begin{bmatrix} (r_0 / 255)^{1/\gamma_r} \\ (g_0 / 255)^{1/\gamma_g} \\ (b_0 / 255)^{1/\gamma_b} \end{bmatrix}$$

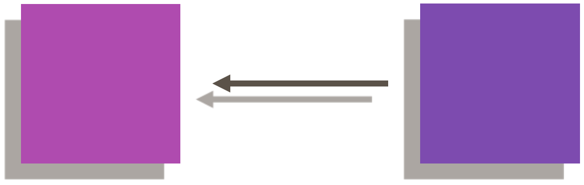
Example of a nonlinear operator: gamma correction

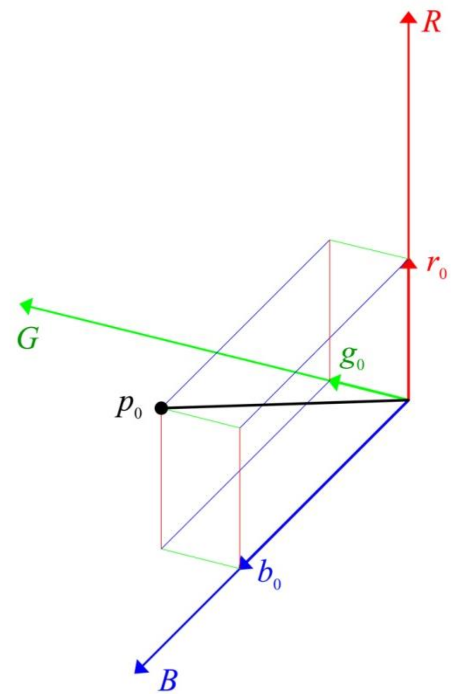
Linear Transformation of Color



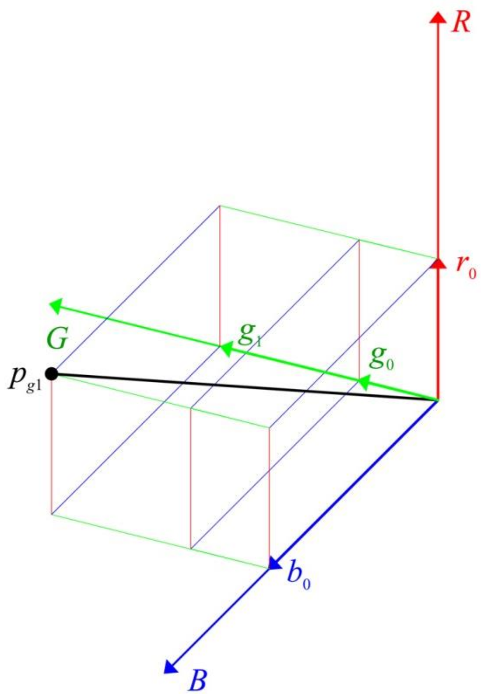
$$\begin{bmatrix} r_1 \\ g_0 \\ b_0 \end{bmatrix} = \begin{bmatrix} r_1/r_0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_0 \\ g_0 \\ b_0 \end{bmatrix}$$

$$\begin{bmatrix} 175 \\ 75 \\ 175 \end{bmatrix} \longleftarrow \begin{bmatrix} 125 \\ 75 \\ 175 \end{bmatrix}$$



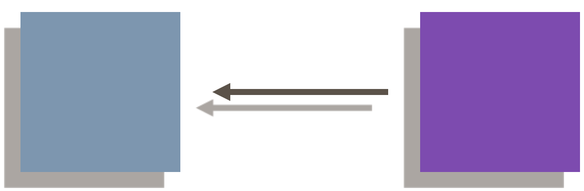


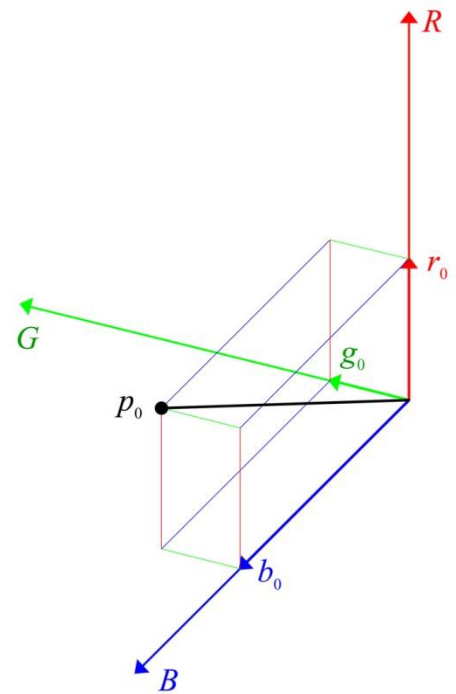
Linear Transformation of Color



$$\begin{bmatrix} r_0 \\ g_1 \\ b_0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & g_1/g_0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_0 \\ g_0 \\ b_0 \end{bmatrix}$$

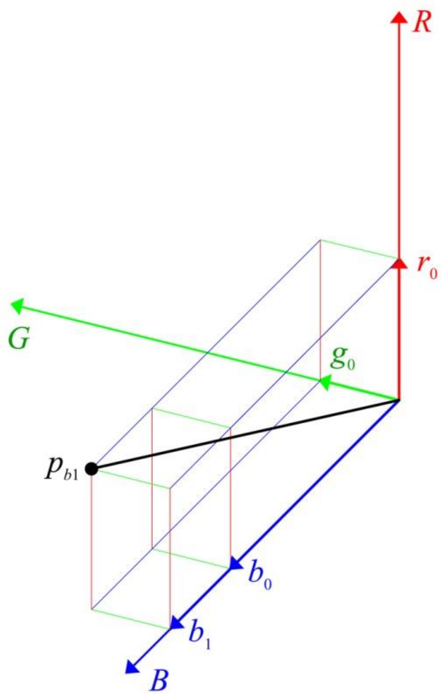
$$\begin{bmatrix} 125 \\ 150 \\ 175 \end{bmatrix} \longleftarrow \begin{bmatrix} 125 \\ 75 \\ 175 \end{bmatrix}$$



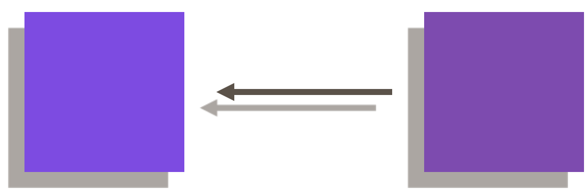


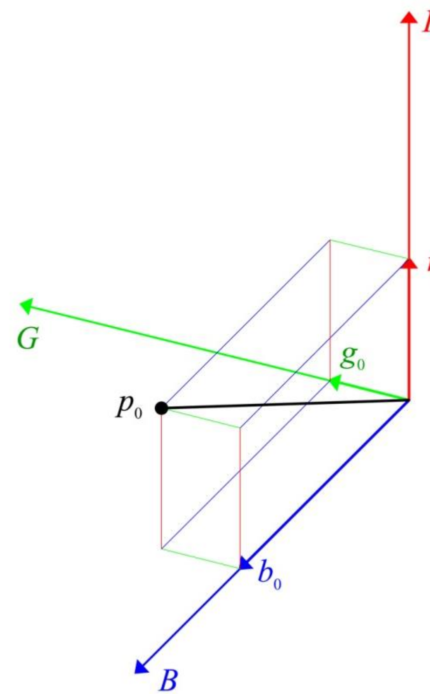
Linear Transformation of Color

$$\begin{bmatrix} r_0 \\ g_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & b_1/b_0 \end{bmatrix} \begin{bmatrix} r_0 \\ g_0 \\ b_0 \end{bmatrix}$$

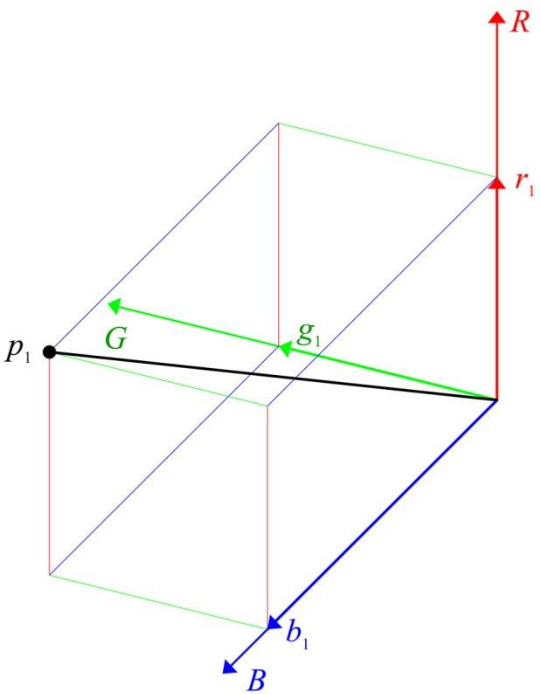


$$\begin{bmatrix} 125 \\ 75 \\ 225 \end{bmatrix} \longleftarrow \begin{bmatrix} 125 \\ 75 \\ 175 \end{bmatrix}$$



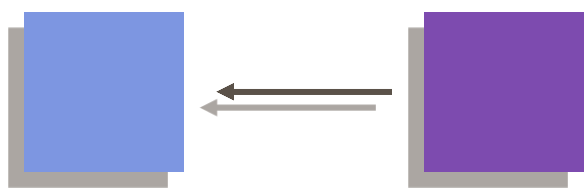


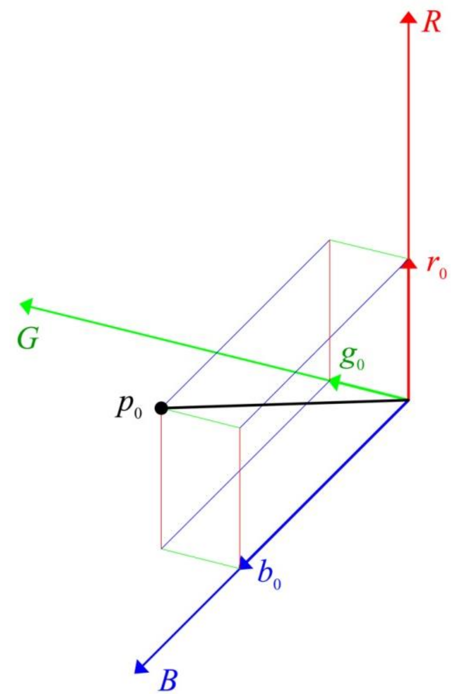
Linear Transformation of Color



$$\begin{bmatrix} r_1 \\ g_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} r_1/r_0 & 0 & 0 \\ 0 & g_1/g_0 & 0 \\ 0 & 0 & b_1/b_0 \end{bmatrix} \begin{bmatrix} r_0 \\ g_0 \\ b_0 \end{bmatrix}$$

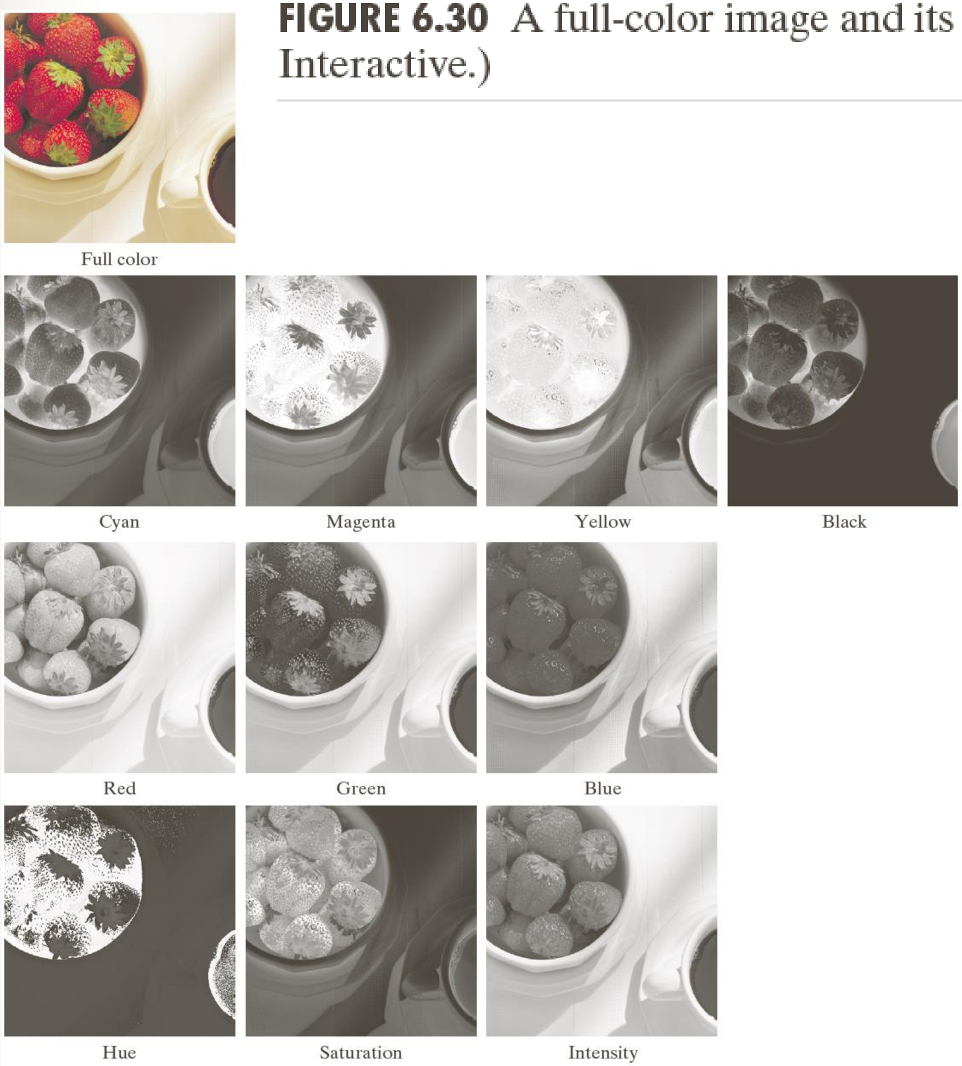
$$\begin{bmatrix} 175 \\ 150 \\ 225 \end{bmatrix} \longleftarrow \begin{bmatrix} 125 \\ 75 \\ 175 \end{bmatrix}$$





Color Transformation

FIGURE 6.30 A full-color image and its various color-space components. Interactive.)



Color Transformation

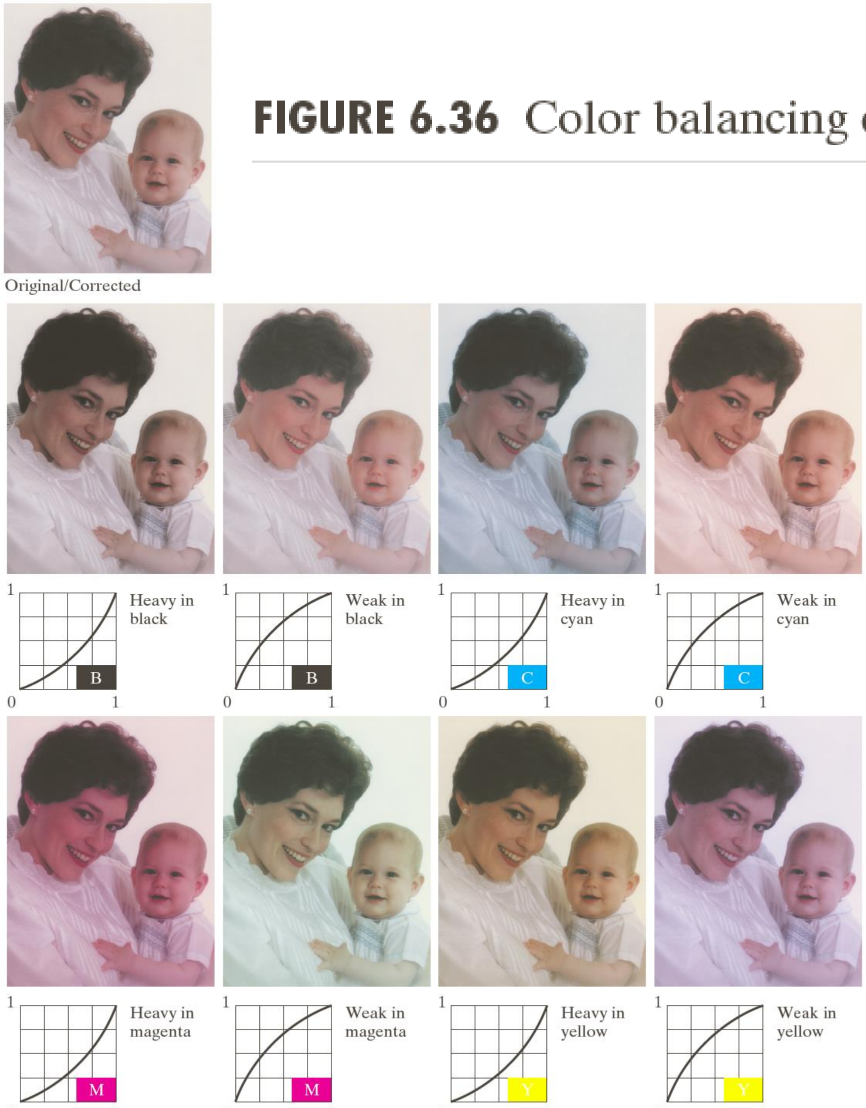


FIGURE 6.36 Color balancing corrections for CMYK color images.

Color Image Example -- RGB

a	b
c	d

FIGURE 6.38
(a) RGB image.
(b) Red component image.
(c) Green component. (d) Blue component.

Color Image Example -- HSI

a b c

FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

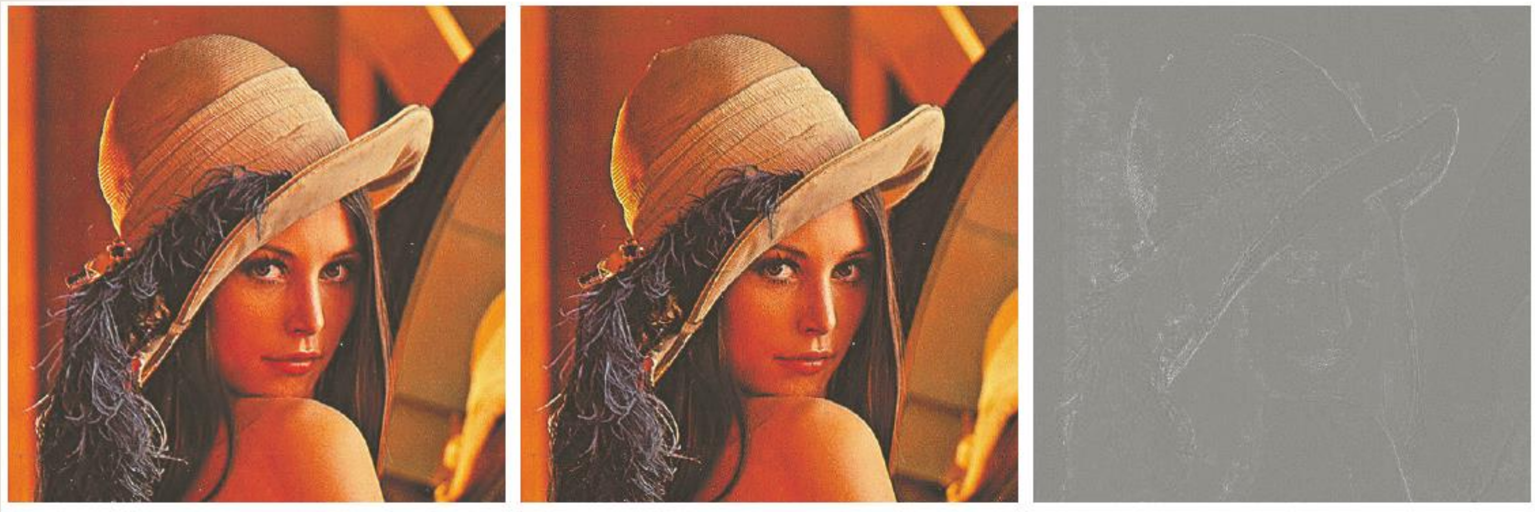
Color Image Smoothing

Average the RGB component vectors in a mask *nxn*

a b c

FIGURE 6.40 Image smoothing with a 5×5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

Color Image Sharpening



a b c

FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

Edge Detection

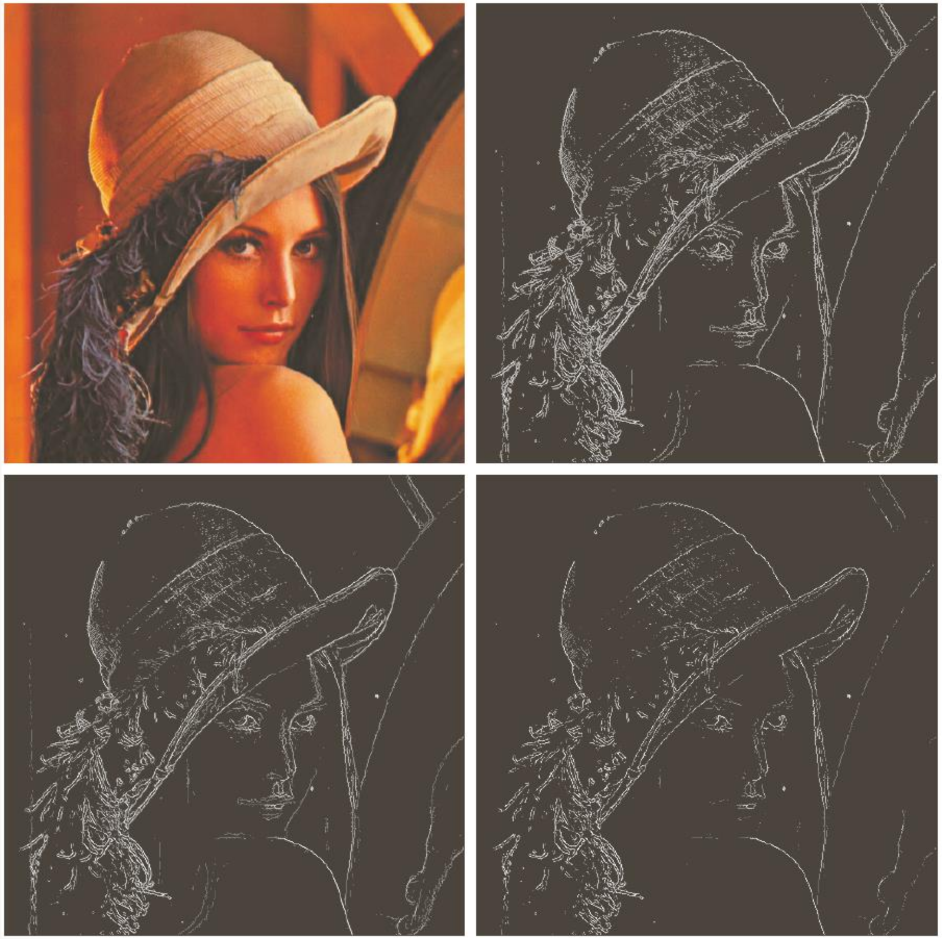


FIGURE 6.46
(a) RGB image.
(b) Gradient computed in RGB color vector space.
(c) Gradients computed on a per-image basis and then added.
(d) Difference between (b) and (c).

Edge Detection

a b c

FIGURE 6.47 Component gradient images of the color image in Fig. 6.46. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.46(c).

HW 1 -- Due: 2/17

- Roy will present HW1 on 2/18.
- Read Chapter 6
- Next class: Image enhancement in spatial domain